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On the potential of recurrent neural networks for modeling path dependent plasticity

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ABSTRACT

The mathematical description of elastoplasticity is a highly complex problem due to the possible change from elastic to elasto-plastic behavior (and vice-versa) as a function of the loading path. Advanced physics-based plasticity models usually feature numerous internal variables (often of tensorial nature) along with a set of evolution equations and complementary conditions. In the present work, an attempt is made to come up with a machine-learning based model that can replicate the predictions anisotropic Yld2000-2d model with homogeneous anisotropic hardening (HAH). For this, a series of modeling problems of increasing complexity is formulated and sequentially addressed using neural network models. It is demonstrated that basic fully-connected neural network models can capture the characteristic non-linearities in the uniaxial stress-strain response such as the Bauschinger effect, permanent softening or latent hardening. A neural network with gated recurrent units (GRUs) and fully-connected layer is proposed for the modeling of plane stress plasticity for arbitrary loading paths. After training and testing the model through comparison with the Yld2000-2d/HAH model, the recurrent neural network model is also used to model the multi-axial stress-strain response of a two-dimensional foam. Here, the comparison with the results from unit cell simulations provided another validation of the proposed data-driven modeling approach.

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1. Introduction

The availability of reliable computational models describing the large deformation behavior of solids is an essential element for the success of new material solutions. Most existing material models for finite element software have been developed using a physics-based approach: first, the governing mechanisms are identified and then described in an approximate manner using a set of algebraic and partial differential equations. Depending on the length scale at which the relevant mech-

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Nomenclature

Abbreviation	Meaning
HAH	Homogeneous Anisotropic Hardening
ANN	Artificial Neural Network
FCNN	Fully-Connected Neural Network
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
FLC	Forming Limit Curve
DD	Diagonal Direction
RD	Rolling Direction
TD	Transverse Direction
RVE	Representative Volume Element
MSE	Mean Squared Error
N_{NPL}	total number of neurons per layer
N_{HL}	total number of hidden layers
N_{IN}	length of the input vector
N_{OUT}	length of the output vector
N_{NPU}	number of neurons per unit
N_{GRU}	number of GRU memory cells
N_{train}	total number of data for training
N_{para}	total number of training parameters

Designations

Symbol	*Description*
$\sigma[t]$	current true stress at the time t
$\varepsilon[t]$	current logarithmic axial strain at the time t
$\mathcal{L}$	total length of the strain path
$\mathcal{L}_\alpha$	total length of the strain path for each loading direction
$\bar{\sigma}$	anisotropic equivalent stress
$k[\bar{\varepsilon}_p]$	deformation resistance
σ_{sat}	ultimate stress
σ_y	initial yield stress
q	crystallographic structure exponent
$\mathbf{x}', \mathbf{x}''$	linear-transformations of the Cauchy stress vector
$\mathbf{L}', \mathbf{L}''$	transformation matrices
$\mathbf{s}$	deviatoric stress tensor
$\phi^q[\mathbf{s}]$	stable component of the anisotropic equivalent stress
$\phi_h^q[\mathbf{s}]$	fluctuating component of the anisotropic equivalent stress
$f_1^q, f_2^q, g_1^{-q}, g_2^{-q}$	auxiliary variables
$\widehat{\mathbf{h}}^s$	microstructure deviator
$\hat{\mathbf{s}}$	normalized stress deviator
σ_f	actual flow stress
$\mathbf{h}^{(i)}$	i-th hidden layer
$\mathbf{b}^{(i)}$	bias vector of the i-th hidden layer
$\mathbf{W}^{(i)}$	weight matrices of the i-th hidden layer
$\mathbf{x}_{<\tau>}, 0 \leq \tau \leq t$	history of the input vector $\mathbf{x}$ (of length N_{IN}) from 0 to time t
$\mathbf{y}_{<t>} = \mathbf{y}[t]$	output vector of length N_{OUT} at time t
$\mathbf{h}^{(i)}_{<t>}$	hidden state at time t
$\mathbf{h}^{(i)}_{<t-1>}$	hidden state at time t-1
$\mathbf{W}_r^{(1)}$	reset gate weight matrix 1
$\mathbf{U}_r^{(1)}$	reset gate weight matrix 2
$\mathbf{b}_r^{(1)}$	reset gate bias vector
$\mathbf{W}_z^{(1)}$	forget gate weight matrix 1
$\mathbf{U}_z^{(1)}$	forget gate weight matrix 2

$\mathbf{b}_z^{(1)}$	forget gate bias vector
ζ	hard-sigmoid function
$\tilde{\mathbf{h}}$	candidate hidden state
$\mathbf{F}$	macroscopic deformation gradient
$\mathbf{P}$	macroscopic first Piola-Kirchhoff stress tensor
ε_1, ε_2	principal strains

anisms take place, multiple levels of homogenization are needed to come up with estimates of the stress-strain response at the macroscopic level.

In sheet metal plasticity, the main ingredients of macroscopic material models are the elastic constitutive equation, the yield function, the flow rule, and a set of evolution equations describing the hardening response. Aside from the basic isotropic Levy-von Mises model, anisotropic yield functions and flow potentials have emerged over the last six decades such as the Hill (1948, 1990) and Hershey families (e.g. Karafillis and Boyce, 1993; Barlat et al., 2003a, 2003b). Furthermore, major developments were concerned with hardening laws including the description of the Bauschinger effect (e.g. Bauschinger, 1886; Prager, 1956; Armstrong et al., 1966; Mollica et al., 2001), kinematic hardening (e.g. Chaboche, 1989), work-hardening stagnation (e.g. Yoshida and Uemori, 2002), permanent softening (e.g. Rauch et al., 2007), latent hardening (e.g. Barlat et al., 2013), distortional hardening (e.g. Barlat et al., 2011, 2014), strain rate effects (e.g. Marsh and Campbell, 1963; Gurrutxaga-Lerma et al., 2015; Balasubramanian and Anand, 2002; Nguyen et al., 2017), and thermal softening (e.g. Stainier et al., 2002; Ling and Belytschko, 2009). For practical applications, the large number of parameters of modern constitutive models are usually calibrated such as to provide an optimal fit for a wide range of experiments (e.g. Abi-Akl and Mohr, 2017; Gorji and Mohr, 2018). In other words, the analysis of the governing physical mechanisms guides the selection of suitable mathematical models, while the model parameters are identified through optimization (minimization of the error between model predictions and experimental results). Given that modern physics-based material models require the use of optimization software for identifying their parameters from experiments (e.g. Gorji et al. 2018; Pack et al., 2018), the question comes up if the direct use of data-driven neural network-based models may provide a viable alternative to physics-based models.

Artificial neural networks (ANNs) are a class of machine learning algorithms that consist of many artificial neurons (often referred to as units) that are arranged in a layered structure. As demonstrated by Cybenko (1989), any smooth non-linear function may be represented by a neural network function with one hidden layer and sigmoidal activation. In mechanics, deep feedforward networks composed of multiple Fully-Connected Neural Network (FCNN) layers have been used as meta-modeling technique to offer alternatives to traditional approaches for structural applications (e.g. Kohar et al., 2016, 2017), material parameter identification (e.g. Haj-Ali et al., 2007), constitutive modeling (e.g. Al-Haik et al., 2006), control manufacturing part attributes such as springback (Cao et al., 1999.), as well as the prediction of the mechanical response of polycrystalline metals (Frankel et al., 2019).

The ANN approach provides a useful framework for describing unconventional hardening behavior observed in tensile experiments. For example, Jenab et al. (2016) employed a shallow neural network to describe the stress-strain response of anisotropic aluminum 5182-O for strain rates ranging from 10^{-3} to 10^{3}/s as a function of the material orientation, the true strain, and the true strain rate. Li et al. (2019) trained an ANN as an integral part of a Johnson-Cook type of law to capture the effects of temperature and strain rate on the hardening of dual phase (DP) steels. Pandya et al. (2019) developed a neural network model to describe the temperature-dependent plasticity and fracture of aluminum 7075 in hot-stamping processes. Gorji and Mohr (2019a) showed the potential of FCNN to characterize the hardening and softening behavior of a metals for temperatures ranging from 25°C to 1250°C and strain rates ranging from 10^{-3}/s to 10^{3}/s. Jordan et al. (2019) used a fully connected network to describe the temperature-dependent viscoplastic hardening function of polypropylene material. It is worth noting that they developed a robot-assisted automated testing system to generate the wealth of experimental data needed for machine-learning based constitutive modeling.

The complexity of the needed neural network models increases when other material model elements such as the yield function and flow rule need to be represented. In addition to the strain hardening, Ali et al. (2019) trained an ANN with 80 neurons per hidden layer to describe the behavior of single crystals subjected to uniaxial tension or shear loading. They chose the current stress and three Eulerian angles as the inputs and trained their network based on the results from crystal plasticity simulations. Greve et al. (2019) showed a substantial computational speedup when using ANN to predict the path-dependent forming limit curve (FLC) instead of Marciniak–Kuczynski (M-K) type of analysis. Lefik and Schrefler (2003) discussed the possibility of incorporating FCNN based constitutive models into an FE code for modeling the non-linear stress-strain response of composites subject to elasto-plastic hysteresis and biaxial loading. Man and Furukawa (2011) developed an ANN-based constitutive model of anisotropic carbon fiber reinforced plastics. Palau et al. (2012) used the von Mises plasticity model to generate training data for monotonic, cyclic, butterfly, and random loading paths. Subsequently, they trained an FCNN model representing incremental plasticity. Gorji and Mohr (2019b) took a similar approach to train the FCNN architecture and to describe the plane stress response of DP780 steel for monotonic loading. Rovinelli et al. (2018a, 2018b) used a semi-supervised machine learning model to predict the crack propagation of polycrystalline materials under fatigue loading conditions. Liu et al (2019) proposed a data-driven multiscale material modeling method, which incorporates ana-

lytical homogenization solutions into a neural network model. Liu and Wu (2019) extended their model for application to 3D heterogeneous materials with multiscale sources of nonlinearity such as particle-reinforced hyperelastic rubbers, polycrystalline materials, and elasto-plastic composites. While all of the above work demonstrates the high potential of neural networks as a data-driven alternative to physics-based models, prediction of the stress-strain response of elasto-plastic solids for arbitrary multi-axial loading paths remains challenging.

The family of the neural network models is constantly growing. In addition to FCNN, Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) models have been developed for applications such as image analysis and speech recognition (e.g. Goodfellow et al., 2017). A recent study by Mozaffar et al. (2019) showed their merits in predicting the behavior of composites subject to nonlinear loading. Even though RNNs have not yet been applied to the modeling of sheet metal, they seem to be particularly suitable for modeling path-dependent plasticity. Unlike FCNNs, recurrent neural networks are equipped with history-dependent internal variables that may potentially mimic the role of objects such as plastic strains and back stresses in physics-based plasticity. Training of RNNs has long been considered a challenge due to vanishing and exploding gradients in the backpropagation process (Hochreiter et al., 2001). However, Long Short-Term Memory (LSTM) models (e.g. Hochreiter and Schmidhuber, 1997) and Gated Recurrent Units (GRU) (e.g. Cho et al., 2014) are examples for successful formulations of RNNs that overcome their notorious training issues.

It is the goal of the present work to construct FCNN- as well as RNN-based constitutive models to describe the large deformation response of elasto-plastic solids. The Homogeneous Anisotropic Hardening (HAH) model with the Yld2000-2d yield function (e.g. Barlat et al., 2011; Barlat et al., 2013, 2014) will serve as a physics-based reference solution, thereby challenging the RNN model to represent complex anisotropic hardening phenomena. To gain insight into the modeling capabilities of neural networks, several constitutive modeling problems of increasing degree of difficulty are treated. Section 2 provides an overview of the corresponding problem statements. In Section 3, we recall briefly the constitutive equations for the HAH model, before detailing the formulations of the fully-connected and recurrent neural network models with gated recurrent units in Section 4. In Section 5, it is then demonstrated that FCNNs and RNNs are able to replicate the predictions of conventional plasticity models, including those of the complex HAH model for an aluminum alloy and a mild steel. A combined RNN-FCNN model is then further challenged in Section 6, where we describe the macroscopic stress-strain response of a two-dimensional metallic foam after training based on the results from unit cell simulations.

2. Problem statements

The primary goal of this work is to demonstrate that conventional physics-based (or at least mechanism-inspired) plasticity models may be substituted by neural network models. As a representative reference solution, an anisotropic plasticity model (Yld2000-2D) with combined isotropic, kinematic and distortional hardening (HAH model) is chosen. Since the application of machine learning in the area of plasticity is still at an early stage of development, we elaborate neural network models for problems of increasing complexity:

- Problem I: Uniaxial stress-strain response for strain-driven loading with loading direction reversal (unloading). We seek for a neural network function f^{I}_{NN} that predicts the current true stress $\sigma[t]$ (at the time t) based on the current logarithmic axial strain $\varepsilon = \varepsilon[t]$, and the total length of the strain path, $\mathcal{L} = \int_0^t |\dot{\varepsilon}[\tau]| d\tau$. Formally, the neural network function to be estimated corresponds to the mapping

$$f^{I}_{NN} : \begin{pmatrix} \varepsilon \\ \mathcal{L} \end{pmatrix} \rightarrow \sigma. \tag{1}$$

- Problem II: Uniaxial stress-strain response for strain-driven loading with loading direction change by an angle α with respect to the monotonic loading direction ($\alpha = 0$). The loading direction reversal considered in Problem I corresponds to a 180° change of the direction of loading. Under problem II, we consider arbitrary loading direction changes from 0 to 90°. The corresponding neural network problem is defined as

$$f^{II}_{NN} : \begin{pmatrix} \varepsilon \\ \alpha \\ \mathcal{L}_\alpha \end{pmatrix} \rightarrow \sigma, \tag{2}$$

with $\mathcal{L}_\alpha$ denoting the length of the strain path for each loading direction.

- Problem III: Multi-axial plane stress response for monotonic loading along radial paths in strain space. Here, the attention is limited to a special class of loading paths which reduces the neural network problem to a mapping from the current strain vector to the current stress vector,

$$f^{III}_{NN} : \begin{pmatrix} \varepsilon_{xx}[t] \\ \varepsilon_{yy}[t] \\ \varepsilon_{xy}[t] \end{pmatrix} \rightarrow \begin{pmatrix} \sigma_{xx}[t] \\ \sigma_{yy}[t] \\ \sigma_{xy}[t] \end{pmatrix}. \tag{3}$$

- Problem IV: Multi-axial plane stress response for arbitrary loading paths in strain space. Different from Problem III, the current stresses can no longer be expressed as a function of the current strains. Instead, they are a function of the loading paths $\varepsilon_i[\tau]$, $0 \leq \tau \leq t$. Formally, the machine learning problem is therefore written as

$$f_{NN}^{IV} : \begin{pmatrix} \varepsilon_{xx}[\tau], 0 \leq \tau \leq t \\ \varepsilon_{yy}[\tau], 0 \leq \tau \leq t \\ \varepsilon_{xy}[\tau], 0 \leq \tau \leq t \end{pmatrix} \rightarrow \begin{pmatrix} \sigma_{xx}[t] \\ \sigma_{yy}[t] \\ \sigma_{xy}[t] \end{pmatrix}. \tag{4}$$

The secondary goal of this work is to demonstrate that machine learning provides also a powerful tool for describing the macroscopic stress-strain response of new materials (for which the development of physics-based constitutive equations is still underway). The type of neural network model developed for Problem IV will thus also be trained and validated based on the results from unit cell analysis for a two-dimensional foam.

3. Physics-based plasticity model

3.1. Constitutive equations

The yield function is written in terms of the anisotropic equivalent stress $\bar{\sigma}$ and the deformation resistance $k[\bar{\varepsilon}_p]$,

$$f = \bar{\sigma} - k[\bar{\varepsilon}_p] = 0. \tag{5}$$

In this work, isotropic hardening is defined by the Hockett-Sherby function (Hockett and Sherby, 1975),

$$k[\bar{\varepsilon}_p] = \sigma_{sat} - (\sigma_{sat} - \sigma_y) \exp\left(-m\bar{\varepsilon}_p^n\right). \tag{6}$$

σ_{sat} and σ_y denote the ultimate stress and the initial yield stress, respectively, whereas m and n control the hardening evolution. The non-quadratic Yld2000-2d yield function (Barlat et al., 2003) is employed to describe the plastic behavior. For this, the anisotropic equivalent stress is defined as,

$$\bar{\sigma}_{Yld2000-2d} = \frac{1}{2^{1/q}} \left(\left| \mathrm{x}_1' - \mathrm{x}_2' \right|^q + \left| 2\mathrm{x}_2'' + \mathrm{x}_1'' \right|^q + \left| 2\mathrm{x}_1'' + \mathrm{x}_2'' \right|^q \right)^{\frac{1}{q}}, \tag{7}$$

where q is a real number between two and infinity related to the crystal structure; the standard recommendation based on the works of Logan and Hosford (1980) is $q = 8$ for FCC- and $q = 6$ for BCC-materials. $\mathbf{x}' = \{\mathrm{x}'_{xx}, \mathrm{x}'_{yy}, \mathrm{x}'_{xy}\}$ and $\mathbf{x}'' = \{\mathrm{x}''_{xx}, \mathrm{x}''_{yy}, \mathrm{x}''_{xy}\}$ denote linear-transformations of the Cauchy stress vector $\boldsymbol{\sigma}$ (whose components correspond to those of the stress tensor written in the material coordinate system), $\mathbf{x}' = \mathbf{L}'\boldsymbol{\sigma}$ and $\mathbf{x}'' = \mathbf{L}''\boldsymbol{\sigma}$ with the transformation matrices,

$$\begin{aligned} \mathbf{L}' &= \frac{1}{3} \begin{bmatrix} 2\alpha_1 & -\alpha_1 & 0 \\ -\alpha_2 & 2\alpha_2 & 0 \\ 0 & 0 & 3\alpha_7 \end{bmatrix} \\ \mathbf{L}'' &= \frac{1}{9} \begin{bmatrix} -2\alpha_3 + 2\alpha_4 + 8\alpha_5 - 2\alpha_6 & \alpha_3 - 4\alpha_4 - 4\alpha_5 + 4\alpha_6 & 0 \\ 4\alpha_3 - 4\alpha_4 - 4\alpha_5 + \alpha_6 & -2\alpha_3 + 8\alpha_4 + 2\alpha_5 - 2\alpha_6 & 0 \\ 0 & 0 & 9\alpha_8 \end{bmatrix}. \end{aligned} \tag{8}$$

Barlat et al. (2013) proposed the phenomenological Homogeneous Anisotropic Hardening (HAH) model to represent the Bauschinger effect without explicitly modeling kinematic hardening along a continuous strain path. The anisotropic equivalent stress $\bar{\sigma}[\mathbf{s}]$ is expressed as a function of the deviatoric stress tensor $\mathbf{s}$. It corresponds to a combination of (i) a stable component $\phi^q[\mathbf{s}]$ that can be any first-order homogeneous yield function, i.e., the Yld2000-2d function in the present work, and (ii) a fluctuating component $\phi_h^q[\mathbf{s}]$,

$$\bar{\sigma}[\mathbf{s}] = \left(\phi^q[\mathbf{s}] + \phi_h^q[\mathbf{s}]\right)^{\frac{1}{q}} = \left(\phi^q[\mathbf{s}] + f_1^q \left|\widehat{\mathbf{h}}^s : \mathbf{s} - \left|\widehat{\mathbf{h}}^s : \mathbf{s}\right|\right|^q + f_2^q \left|\widehat{\mathbf{h}}^s : \mathbf{s} + \left|\widehat{\mathbf{h}}^s : \mathbf{s}\right|\right|^q\right)^{\frac{1}{q}}, \tag{9}$$

where $f_1^q = (g_1^{-q} - 1)^{-q}$ and $f_2^q = (g_2^{-q} - 1)^{-q}$ are two auxiliary variables; Note that there is no contribution of the fluctuating component $\phi_h^q[\mathbf{s}]$ to the first tensile stage, which leads to the parameters $f_1 = f_2 = 0$ and consequently $g_1 = g_2 = 0$. The term $\phi_h^q[\mathbf{s}]$ is written as a function of a microstructure deviator $\widehat{\mathbf{h}}^s$. The deformation history is memorized in this second-order tensor,

$$\widehat{\mathbf{h}}^s = \frac{\mathbf{h}^s}{\sqrt{\frac{8}{3}\mathbf{h}^s : \mathbf{h}^s}}. \tag{10}$$

Table 1
The yield stresses and average Lankford ratios of the sheet samples.

	σ^y_{RD}[MPa]	σ^y_{DD}[MPa]	σ^y_{TD}[MPa]	σ^y_b[MPa]	R_{RD} [-]	R_{DD} [-]	R_{TD} [-]	R_b [-]
AA5182	147.7	144.9	146.4	130.0	0.72	0.74	0.71	1.13
DC05	171.0	179.9	177.3	195.0	2.00	1.47	2.52	0.85

The tensor $\widehat{\mathbf{h}}^s$corresponds to the stress deviator at the first plastic deformation ($\widehat{\mathbf{h}}^s = \mathbf{s}$). As the loading direction changes ($\widehat{\mathbf{h}}^s \neq \mathbf{s}$), the relation between the microstructure deviator and the deviatoric stress is continuously modified,

$$
\begin{aligned}
&\text{If} \widehat{\mathbf{h}}^s : \mathbf{s} \geq 0 && \text{If} \widehat{\mathbf{h}}^s : \mathbf{s} < 0 \\
&\frac{dg_1}{d\bar{\varepsilon}} = k_2\left(k_3\frac{\bar{\sigma}(0)}{\bar{\sigma}(\bar{\varepsilon})} - g_1\right) && \frac{dg_1}{d\bar{\varepsilon}} = k_1\left(\frac{g_4 - g_1}{g_1}\right) \\
&\frac{dg_2}{d\bar{\varepsilon}} = k_1\left(\frac{g_3 - g_2}{g_2}\right) && \frac{dg_2}{d\bar{\varepsilon}} = k_2\left(k_3\frac{\bar{\sigma}(0)}{\bar{\sigma}(\bar{\varepsilon})} - g_2\right) \\
&\frac{dg_4}{d\bar{\varepsilon}} = k_5(k_4 - g_4) && \frac{dg_3}{d\bar{\varepsilon}} = k_5(k_4 - g_3) \\
&\frac{d\widehat{\mathbf{h}}^s}{d\bar{\varepsilon}} = k\left(\widehat{\mathbf{s}} - \frac{8}{3}\widehat{\mathbf{h}}\left(\widehat{\mathbf{h}}^s : \widehat{\mathbf{s}}\right)\right) && \frac{d\widehat{\mathbf{h}}^s}{d\bar{\varepsilon}} = k\left(-\widehat{\mathbf{s}} + \frac{8}{3}\widehat{\mathbf{h}}\left(\widehat{\mathbf{h}}^s : \widehat{\mathbf{s}}\right)\right)
\end{aligned}
\quad , \tag{11}
$$

where $\hat{\mathbf{s}}$ is the normalized stress deviator, $\hat{\mathbf{s}} = \mathbf{s}/\sqrt{\frac{8}{3}\mathbf{s} : \mathbf{s}}$. Recall that the auxiliary variables g_1, g_2, $g_3$and g_4 all control the shape and size of the yield surface (Eq. 9). As pointed out by Barlat et al. (2013), it can be shown that the differential equations for g_3 and g_4 are particularly important for modeling the permanent softening after loading direction reversal; it is detected when the sign of $\widehat{\mathbf{h}}^s : \mathbf{s}$ changes. The permanent softening is permitted when $k_4 = 1$ or $k_5 = 1$ and $g_3 = g_4 = 1$.

The HAH model is extended from its original version to account for latent effects (Barlat et al., 2013) that are responsible for (i) stress overshooting after the subsequent second strain path, and (ii) a lower re-yielding stress level. The extended HAH captures latent hardening by scaling the actual flow stress σ_f with the factor g_L,

$$
\frac{dg_L}{d\bar{\varepsilon}} = k_L\left(\frac{\sigma_f - \sigma_y}{\sigma_f}\left(\sqrt{L\left(1 - \cos^2\chi\right) + \cos^2\chi} - 1\right) + 1 - g_L\right), \tag{12}
$$

where g_L is equal to unity under monotonic deformation. It evolves, accordingly, by changing the loading direction. The stress evolution depends on the parameters k_L, L, and $\cos\chi = \frac{8}{3}\mathbf{s} : \hat{\mathrm{h}}$. Note that $\cos\chi$ becomes -1, 0, and 1 for loading direction reversal, orthogonal path changes, and monotonic loadings, respectively.

Using an analogous approach, the function g_C is proposed to capture the latent contraction. It scales the monotonic flow stress under strain path changes,

$$
g_C = \frac{1}{f_C + \cos^2\chi\,(1 - f_C)}, \tag{13}
$$

whereas the variable f_C evolves according to,

$$
\frac{df_C}{d\bar{\varepsilon}} = k_C\left((C - 1)\cos^2\chi + 1 - f_C\right). \tag{14}
$$

with the two material constants k_C and C. By continuing the deformation at the second stage, the microstructure deviator $\hat{\mathrm{h}}^s$rotates towards the deviatoric stress $\hat{\mathbf{s}}$, which leads to $\cos^2\chi = 1$; consequently g_L or g_C attain unity again.

3.2. Material model parameters for aluminum alloy 5182

The aluminum alloy 5182 is chosen as the first material for the model developments. The parameters of the Yld2000-2d/HAH plasticity model are obtained from Peters (2015). The measured directional stresses and the corresponding anisotropic parameters are listed in Table 1. The Lankford parameters (*R*-values), defined as the ratio of the plastic width-to-thickness strain in a uniaxial tension experiment, $R = d\varepsilon^p_{width}/d\varepsilon^p_{thickness}$, are also used to characterize the material anisotropy. Fig. 1a and b show the continuous variation of the calculated yield stress and *R*-values of the Yld2000-2d model with respect to the rolling direction of the material; solid circles show selected experimental results (for the rolling, diagonal and transverse directions, R_{RD},R_{DD}, and R_{TD}). The lowest yield stress occurs at an angle 45° with respect to the rolling direction ($\sigma_{DD} = 144.9$MPa) whereas the yield stresses for the rolling and transverse directions are almost identical ($\sigma_{RD} = 147.7$MPa,$\sigma_{TD} = 146.4$MPa). The plastic anisotropy of this rolled sheet is relatively uniform; the minimum R-value ($R_{RD} = 0.72$) is 2% lower than the maximum one ($R_{DD} = 0.74$). Fig. 1c shows the shear stress-strain response with loading direction reversal after logarithmic shear strains of 0.05, 0.10, and 0.15 as predicted by the HAH plasticity model. The Bauschinger effect and the permanent softening after the loading direction reversal can be clearly seen in this figure.

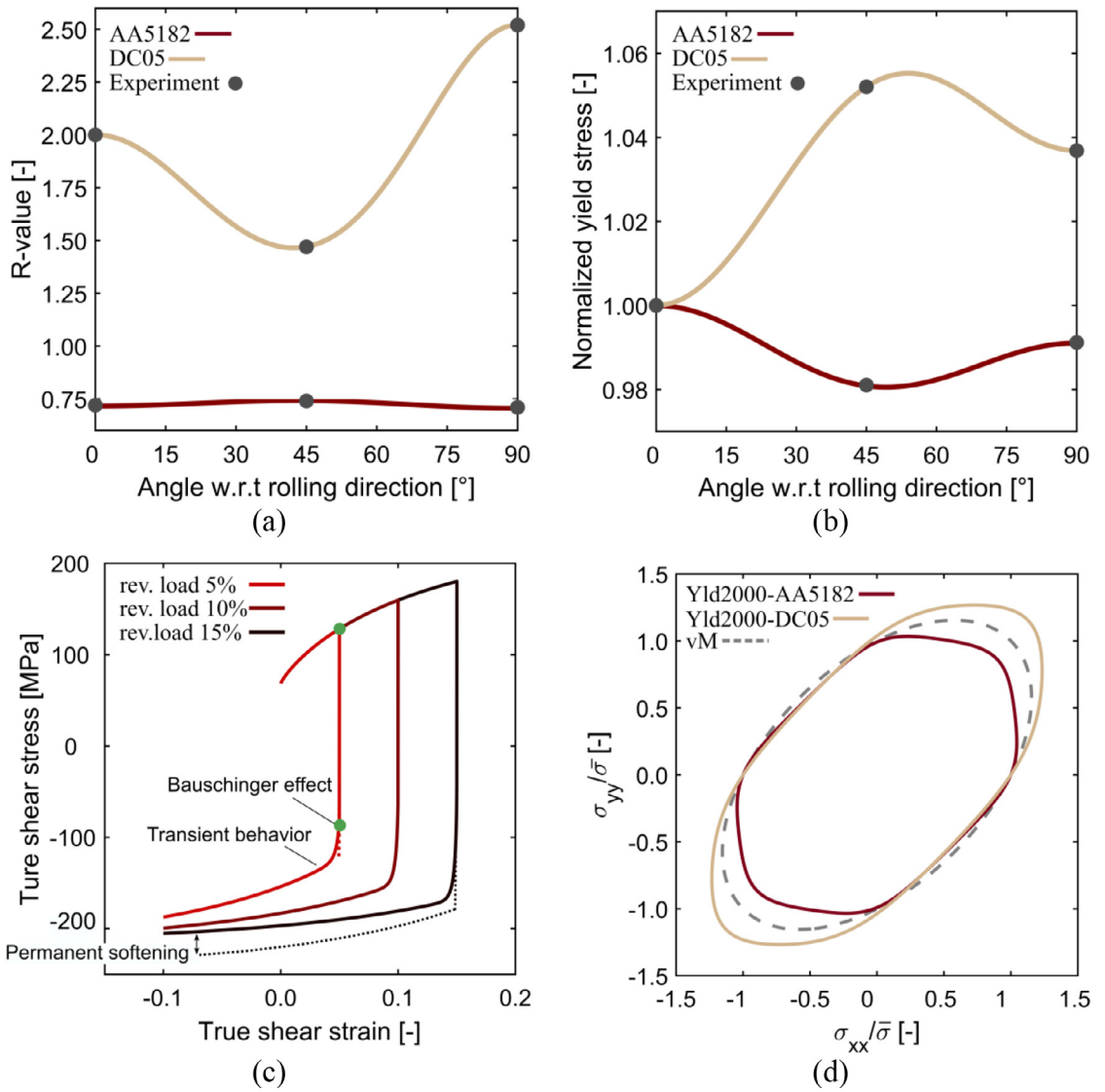


Fig. 1. (a) Lankford coefficient, and (b) yield stress of AA5182 and DC05 as a function of the material orientation with respect to the rolling direction; (c) shear stress-strain response as predicted by the calibrated HAH model for AA5182 (solid lines); (d) yield loci for AA5182 (red) and DC05 steel (yellow); the dashed envelope depicts the von Mises yield locus. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

3.3. Material model parameters for DC05 steel

The deep drawing steel DC05 (also known as 1.0312) is selected as the second demonstrator material. The plasticity parameters of this alloy are also obtained from Peters (2015). Table 1 summarizes the mechanical properties of DC05 as obtained from uni-axial and equi-biaxial tension experiments. As compared to AA5182, anisotropic effects are more pronounced in DC05 steel. The highest yield stress of the material is observed for uniaxial tension at an angle of 54° with respect to the rolling direction ($\sigma_{54} = 180.4$MPa); the yield stress for loading along the transverse direction is almost 4% higher than that for the rolling direction ($\sigma_{\text{RD}} = 171.0$MPa,$\sigma_{\text{TD}} = 177.3$MPa). The lowest R-value of 1.47 is measured for the diagonal direction, which is 36% and 71% lower than that for the rolling and transverse directions, respectively (see Figs. 1a and b). Fig. 1d depicts the normalized non-quadratic Yld2000-2d loci of both aluminum 5182 and DC05 steel. For reference, the isotropic von Mises plasticity yield locus is shown as a gray dashed envelope. The shape of the yield envelope of the DC05 steel deviates significantly from that for the aluminum 5182 between the uniaxial and the equi-biaxial loading cases. The vertices of the yield envelope are less pronounced for the DC05 material ($q = 6$ versus $q = 8$ for AA5182).

4. Neural network model formulations

Neural networks are used to describe the relationship between the input and output variables for the constitutive modeling problems listed in Section 2. Today, there exists a wealth of different types of neural networks (e.g. van Veen, 2016) among which two distinct types are applied in the present study. Firstly, when estimating the non-linear functional relationship between an input and an output vector (problems I to III), Fully-Connected Neural Networks (FCNNs) are employed. Secondly, when dealing with problems where the output vector at a time t depends on the history of the input vector from 0 to time t (problem IV), so-called Gated Recurrent Unit (GRU) neural networks (Cho et al., 2014) are adopted. Among different recurrent neural network types, the GRUs are retained since they are easier to implement than Long Short-Term Memory (LSTM) units (Hochreiter and Schmidhuber, 1997).

4.1. Fully-connected neural networks

The problem statement for the FCNNs reads

$$f_{FCNN} : \mathbf{x} \rightarrow \mathbf{y}, \tag{15}$$

with $\mathbf{x}$ denoting the input vector of length N_{IN}, and $\mathbf{y}$ denoting the output vector of lengthN_{OUT}. As schematically shown in Fig. 2a, the main building blocks of FCNNs are hidden layers. They are numbered from $i = 1, ..., N_{HL}$, with N_{HL} denoting the total number of hidden layers. The mathematical object describing the ith hidden layer is a vector $\mathbf{h}^{(i)}$. Its length is given by the number of neurons per layer, N_{NPL}, which is assumed to be the same for all N_{HL} hidden layers in the context of this study. The feedforward procedure involves the step-by-step computation of the vectors $\mathbf{x} \rightarrow \mathbf{h}^{(1)} \rightarrow \mathbf{h}^{(2)} \rightarrow ... \rightarrow \mathbf{h}^{(N_{HL})} \rightarrow \mathbf{y}$:

1 Computation of the first hidden layer,

$$\mathbf{h}^{(1)} = \tanh\left[\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}\right] \tag{16}$$

where the $N_{NPL} \times N_{IN}$ matrix $\mathbf{W}^{(1)}$ summarizes the weights, and the N_{NPL}-dimensional vector $\mathbf{b}^{(1)}$ summarizes the bias parameters of the first layer.

1 Computation of the subsequent hidden layers

$$\mathbf{h}^{(i)} = \tanh\left[\mathbf{W}^{(i)}\mathbf{h}^{(i-1)} + \mathbf{b}^{(i)}\right] \text{ with } i = 2, .., N_{HL} \tag{17}$$

Here, the weight matrices $\mathbf{W}^{(i)}$ have the dimension $N_{NPL} \times N_{NPL}$ and the bias vectors $\mathbf{b}^{(i)}$ have the length N_{NPL}.

1 Computation of the output variable

$$\mathbf{y} = \mathbf{W}^{(N_{HL}+1)}\mathbf{h}^{(N_{HL})} + \mathbf{b}^{(N_{HL}+1)} \tag{18}$$

with the $N_{OUT} \times N_{NPL}$ weight matrix $\mathbf{W}^{(N_{HL}+1)}$ and the N_{OUT}-dimensional bias vector $\mathbf{b}^{(N_{OUT})}$.

The total of free parameters of a fully-connected neural network with the hyperparameters $\{N_{HL}, N_{NPL}\}$ is:

$$N_{para} = (N_{IN} + 1)N_{NPL} + (N_{HL} - 1)(N_{NPL} + 1)N_{NPL} + (N_{NPL} + 1)N_{OUT} \tag{19}$$

4.2. Recurrent neural networks

The problem statement for the recurrent neural networks reads

$$f_{RNN} : \mathbf{x}_{<\tau>}, 0 \leq \tau \leq t \rightarrow \mathbf{y}_{<t>}, \tag{20}$$

with $\mathbf{x}_{<\tau>}, 0 \leq \tau \leq t$ denoting the history of the input vector $\mathbf{x}$ (of length N_{IN}) from 0 to time t, and $\mathbf{y}_{<t>} = \mathbf{y}[t]$ denoting the output vector of length N_{OUT} at time t. Here, we consider GRU-FCNN networks, i.e. neural networks that contain Gated Recurrent Units (GRUs) in addition to fully-connected hidden layers (Fig. 2b). In close analogy with the above definitions for the hidden layers of FCNNs, a *hidden state* vector $\mathbf{h}^{(i)}$ of length N_{NPU}(number of neurons per unit) is associated with each GRU. However, different from the hidden layers of an FCNN, the GRUs are equipped with memory, i.e. the hidden state $\mathbf{h}^{(i)}_{<t>}$ at time t is not only a function of the current input $\mathbf{x}_{<t>}$, but it depends also on the hidden state $\mathbf{h}^{(i)}_{<t-1>}$ at time t-1 (previous time step). Formally, we write the recurrent relation

$$\mathbf{h}^{(i)}_{<t>} = f\left[\mathbf{h}^{(i)}_{<t-1>}, \mathbf{x}_{<t>}\right]. \tag{21}$$

Overall, the evaluation of a GRU-FCNN network withN_{GRU} gated recurrent units, and N_{HL} hidden layers involves the following step-by-step computation:

$$\mathbf{x}_{<t>} \rightarrow \underbrace{\begin{matrix}\mathbf{h}^{(1)}_{<t-1>} & \mathbf{h}^{(2)}_{<t-1>} & & \mathbf{h}^{(N_{GRU})}_{<t-1>} \\ \downarrow & \downarrow & & \downarrow \\ \mathbf{h}^{(1)}_{<t>} & \rightarrow \mathbf{h}^{(2)}_{<t>} & \rightarrow ... \rightarrow & \mathbf{h}^{(N_{GRU})}_{<t>}\end{matrix}}_{GRU\,network} \underbrace{\rightarrow \mathbf{h}^{(N_{GRU}+1)} \rightarrow \mathbf{h}^{(N_{GRU}+2)} \rightarrow ... \rightarrow \mathbf{h}^{(N_{GRU}+N_{HL})}}_{\text{fully-connected}\,network} \rightarrow \mathbf{y}_{<t>}$$

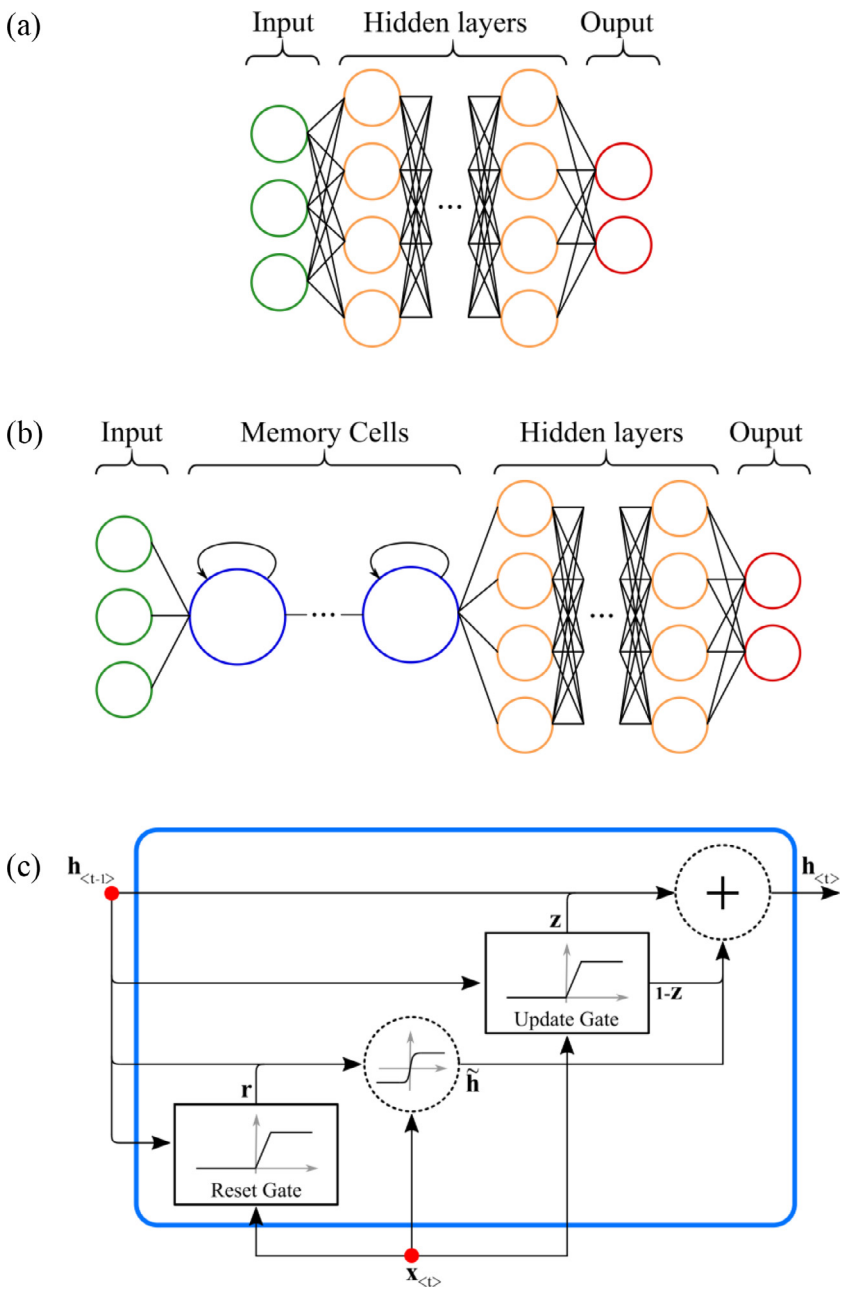


Fig. 2. The main building blocks of (a) fully-connected neural network (FCNN), (b) gated recurrent units (GRUs) followed by fully-connected hidden layers (GRU-FCNN). (c) operations of a GRU cell for updating the hidden state at time t based on the hidden state at time t-1 and the network input at time t.

Specifically, the mathematical operations to be performed in sequence for a GRU-FCNN model with the hyperparameters $\{N_{GRU}, N_{NPU}, N_{HL}, N_{NPL}\}$ read:

1 Computation of the hidden state vector $\mathbf{h}^{(1)}_{<t>}$ for the first GRU at time t. This involves:
  a Computation of the reset gate (vector of length N_{NPU})

$$\mathbf{r}^{(1)} = \varsigma\left[\mathbf{W}_r^{(1)}\mathbf{x}_{<t>} + \mathbf{U}_r^{(1)}\mathbf{h}^{(1)}_{<t-1>} + \mathbf{b}_r^{(1)}\right], \tag{22}$$

with the reset gate weight matrices $\mathbf{W}_r^{(1)}$ (input connection of size $N_{NPU} \times N_{IN}$) and $\mathbf{U}_r^{(1)}$(recurrent connection of size $N_{NPU} \times N_{NPU}$), and the bias vector $\mathbf{b}_r^{(1)}$ of length N_{NPU}. The function $\varsigma[..]$ refers to the hard-sigmoid function which is executed on a per-component basis. Note that the initial hidden state is described by the null matrix $\mathbf{h}^{(1)}_{<0>} = 0$. The schematic layout of the memory GRU cell is drawn in Fig. 2c.

b Activation of the update gate (also called forget gate). In close analogy to the reset gate, the forget gate reads

$$\mathbf{z}^{(1)} = \varsigma\left[\mathbf{W}_z^{(1)}\mathbf{x}_{<t>} + \mathbf{U}_z^{(1)}\mathbf{h}_{<t-1>}^{(1)} + \mathbf{b}_z^{(1)}\right], \tag{23}$$

with the parameters $\mathbf{W}_z^{(1)}(N_{NPU} \times N_{IN}\text{matrix})$, $\mathbf{U}_z^{(1)}$ $(N_{NPU} \times N_{NPU}\text{matrix})$ and $\mathbf{b}_z^{(1)}$.

c Activation of the *candidate hidden state* $\hat{\mathbf{h}}^{(1)}$, which is computed based on the current input $\mathbf{x}_{<t>}$, the previous hidden state $\mathbf{h}_{<t-1>}^{(1)}$, and the output $\mathbf{r}^{(1)}$ of the reset gate,

$$\tilde{\mathbf{h}}^{(1)} = \tanh\left[\tilde{\mathbf{W}}^{(1)}\mathbf{x}_{<t>} + \tilde{\mathbf{U}}^{(1)}\left(\mathbf{r}^{(1)} \circ \mathbf{h}_{<t-1>}^{(1)}\right) + \tilde{\mathbf{b}}^{(1)}\right] \tag{24}$$

where the Hadamard product $\circ$ is introduced to denote the component-by-component of multiplication for creating a vector from two vectors of equal length. Here, the parameters to be learned are stored in the matrices $\tilde{\mathbf{W}}^{(1)}(N_{NPU} \times N_{IN}\text{matrix})$, $\tilde{\mathbf{U}}^{(1)}$ $(N_{NPU} \times N_{NPU}\text{matrix})$ and $\mathbf{b}_z^{(1)}$.

d Computation of the *current hidden state* $\mathbf{h}_{<t>}^{(1)}$ (without introducing any new parameters),

$$\mathbf{h}_{<t>}^{(1)} = \mathbf{z}^{(1)} \circ \mathbf{h}_{<t-1>}^{(1)} + (\mathbf{1} - \mathbf{z}^{(1)}) \circ \tilde{\mathbf{h}}^{(1)} \tag{25}$$

2 Computation of hidden state variables $\mathbf{h}_{<t>}^{(i)}$ for subsequent GRUs $(i = 2, .., N_{GRU})$:

a Reset gates with parameters $\mathbf{W}_r^{(i)}$, $\mathbf{U}_r^{(i)}$ and $\mathbf{b}_r^{(i)}$,

$$\mathbf{r}^{(i)} = \varsigma\left[\mathbf{W}_r^{(i)}\mathbf{h}_{<t>}^{(i-1)} + \mathbf{U}_r^{(i)}\mathbf{h}_{<t-1>}^{(i)} + \mathbf{b}_r^{(i)}\right], \tag{26}$$

b Update gates with parameters $\mathbf{W}_z^{(i)}$, $\mathbf{U}_z^{(i)}$ and $\mathbf{b}_z^{(i)}$,

$$\mathbf{z}^{(i)} = \varsigma\left[\mathbf{W}_z^{(i)}\mathbf{h}_{<t>}^{(i)} + \mathbf{U}_z^{(i)}\mathbf{h}_{<t-1>}^{(i-1)} + \mathbf{b}_z^{(i)}\right], \tag{27}$$

c Candidate hidden states with parameters $\tilde{\mathbf{W}}^{(i)}$, $\tilde{\mathbf{U}}^{(i)}$ and $\tilde{\mathbf{b}}^{(i)}$,

$$\tilde{\mathbf{h}}^{(i)} = \tanh\left[\tilde{\mathbf{W}}^{(i)}\mathbf{h}_{<t>}^{(i-1)} + \tilde{\mathbf{U}}^{(i)}\left(\mathbf{r}^{(i)} \circ \mathbf{h}_{<t-1>}^{(i)}\right) + \tilde{\mathbf{b}}^{(i)}\right]. \tag{28}$$

d Current hidden states:

$$\mathbf{h}_{<t>}^{(i)} = \mathbf{z}^{(i)} \circ \mathbf{h}_{<t-1>}^{(i)} + (\mathbf{1} - \mathbf{z}^{(i)}) \circ \tilde{\mathbf{h}}^{(i)} \tag{29}$$

3 Computation of first hidden layer $\mathbf{h}^{(N_{GRU}+1)}$ of FCNN:

$$\mathbf{h}^{(N_{GRU}+1)} = \tanh\left[\mathbf{W}^{(N_{GRU}+1)}\mathbf{h}_{<t>}^{(N_{GRU})} + \mathbf{b}^{(N_{GRU}+1)}\right]. \tag{30}$$

with the $N_{NPU} \times N_{NPL}$ weight matrix $\mathbf{W}^{(N_{GRU}+1)}$ and the $N_{NPL}-$dimensional bias vector $\mathbf{b}^{(N_{GRU}+1)}$.

4 Computation of subsequent hidden layers $\mathbf{h}^{(i)}(i = N_{GRU} + 1, .., N_{GRU}+N_{HL})$ of the FCNN:

$$\mathbf{h}^{(i)} = \tanh\left[\mathbf{W}^{(i)}\mathbf{h}^{(i-1)} + \mathbf{b}^{(i)}\right]. \tag{31}$$

with the $N_{NPL} \times N_{NPL}$ weight matrices $\mathbf{W}^{(i)}$ and the $N_{NPL}-$dimensional bias vectors $\mathbf{b}^{(i)}$.

5 Computation of the network output:

$$\mathbf{y}_{<t>} = \mathbf{W}^{(N_{GRU}+N_{HL}+1)}\mathbf{h}^{(N_{GRU}+N_{HL})} + \mathbf{b}^{(N_{GRU}+N_{HL}+1)} \tag{32}$$

with the $N_{OUT} \times N_{NPL}$ weight matrix $\mathbf{W}^{(N_{GRU}+N_{HL}+1)}$ and the $N_{OUT}-$dimensional bias vector $\mathbf{b}^{(N_{GRU}+N_{HL}+1)}$.

The total number of parameters N_{para} to be learned in a GRU-FCNN model with the hyperparameters $\{N_{GRU}, N_{NPU}, N_{HL}, N_{NPL}\}$ is

$$\begin{aligned} N_{para} &= 3(N_{IN} + N_{NPU} + 1)N_{NPU} + 3(N_{GRU} - 1)(2N_{NPU} + 1)N_{NPU} + \\ &(N_{NPU} + 1)N_{NPL} + (N_{HL} - 1)(N_{NPL} + 1)N_{NPL} + (N_{NPL} + 1)N_{OUT} \end{aligned} \tag{33}$$

4.3. Model parameter identification (supervised learning)

The weights and biases of the neural network models are determined through supervised learning, i.e. the minimization of the difference between the model predictions and the desired model output. For this, the Mean Squared Error (MSE) is defined,

$$MSE[\mathbf{p}] = \frac{1}{N}\sum_{j=1}^{N}\left(\hat{\mathbf{y}}^{(j)} - \mathbf{y}^{(j)}[\mathbf{p}]\right)^2, \tag{34}$$

with $\mathbf{y}^{(j)} = \mathbf{y}^{(j)}[\mathbf{p}]$ denoting the network outputs for a given set of model parameters $\mathbf{p}$ (with $length[\mathbf{p}] = N_{para}$), and $\hat{\mathbf{y}}^{(j)}$ denoting the desired targets. The data points are numbered with $j = 1, .., N$, with N denoting the total number of training, verification or testing data (to be specified).

In the case of the FCNN models, the parameter identification is performed using the Deep Learning toolbox of Matlab. A Levenberg-Marquardt algorithm with Bayesian regularization is executed to perform the minimization. To facilitate the optimization procedure, the components of the input and output vectors are linearly projected onto the intervals $[-1, +1]$, e.g.

$$\bar{x}_i = -1 + 2\frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \text{ with } i = 1, .., N_{IN} \tag{35}$$

with

$$x_{\min} = \min_{0\le\tau\le t} x_i[\tau] \text{ and } x_{\max} = \max_{0\le\tau\le t} x_i[\tau]. \tag{36}$$

The GRU-FCNN model is trained using the adaptive learning rate optimization algorithm ADAM from the Python deep learning library Keras Chollet, 2015). Here, the network inputs are normalized such that ((1) the variance of each component $\bar{x}_i$ equals to one, and (2) its mean equals to zero (for the training data).

5. Results

5.1. Problem I: uniaxial tension with loading direction reversal

The uniaxial stress-strain response of an HAH material (i.e., a material obeying the constitutive equations outlined in Section 3) is described through a fully-connected neural network. Training data of the form $\{\varepsilon_i, \mathcal{L}_i, \sigma_i\}$ is generated through single element simulations with the parameters for AA5182. Simulations are performed with tensile loading up to a strain of 0.05, 0.10, and 0.15, respectively, followed by reverse loading up to a logarithmic strain of -0.1 (Fig. 1c). From each stress-strain curve (which comprises loading and unloading), 2000 data points are extracted for uniformly spaced strain increments. The total number of data for training thus was $N_{train} = 6000$. All networks are trained using a Levenberg-Marquardt based backpropagation algorithm for up to 1000 epochs assuming a 85-15-0 data split for training, validation, and testing. Note that the testing data is set to zero as the model testing will be performed separately from training by evaluating the model's predictive capabilities for loading paths that are different from those used for generating the training data.

To identify a suitable number of hidden layers N_{HL} and the number of neurons per layer N_{NPL}, a parametric study is performed. The number of hidden layers is varied from 1 to 3, i.e. $\text{N}_{HL} = \{1, 2, 3\}$. Furthermore, the number of neurons per hidden layer is varied from 1 to 30, i.e. $\text{N}_{NPL} = \{1, 2, .., 30\}$. In sum, we thus considered 90 distinct network architectures. For each architecture $\{\text{N}_{HL}, \text{N}_{NPL}\}_i$, the initial weights and biases are randomly assigned using the Nguyen-Widrow initialization function. Table 6a shows the number of free parameters (weights and biases) of the FCNN model as a function of the number of hidden layers N_{HL} (table rows) and neurons per layer N_{NPL} (table columns). It increases from 5 for [2-1-1] network to 1981 for a [2-30-30-30-1] network (i.e. a network with two input variables, three hidden layers with thirty neurons per layer, one output variable). To gain insight into the effect of the number of training runs the corresponding MSEs are shown in Tables 6b-s and plotted in Fig. 3a-c. The first (Table 6b and Fig. 3a) displays the MSEs obtained after one training run per network architecture. The second (Table 6c, Fig. 3c) and third (Table 6d, Fig. 3d) sets of tables and figures show the smallest MSEs obtained (per architecture) after five and ten training runs, respectively. As expected from the standard machine learning literature, there is merit in repeating the training. For instance, the MSE of the network with a [2-30-30-30-1] architecture improves by 12% and 35%, respectively, when increasing the number of training runs to 5 and 10, respectively. Note from Fig. 3 that the MSE of the networks with two and three hidden layers stops improving when increasing the number of neurons per layer from 6 to 30. A plot of the corresponding stress-strain curves (Fig. 4) reveals that for MSEs lower than 10^{-2} MPa^2, the curves predicted by the neural network models lie on top of those provided for training by the HAH model (gray curves in Fig. 4). Among the models considered, the [2-6-6-1] network is the smallest model (as far as the number of parameters in concerned) which provides an MSE well below 10^{-2} MPa^2.

For model testing, simulations with both the HAH and FCNN models are also performed for strain histories (i) with reversal strains that fall into the domain of training (from 0.05 to 0.15), and cases where the reversal strain is (ii) 0.025 (lower), and (iii) 0.175 (higher). The [2-6-6-1] network predictions agree well with the HAH model for the loading scenarios which lie within the range of training data. It provides also a reasonable prediction for a reverse strain of 0.175, but the FCNN model is less accurate when the loading direction is reversed at a strain of 0.025. The three-layer models ([2-25-25-25-1] and [2-30-30-30-1] networks) also perform well within the range of training, while their predictions are less accurate when the models are tested outside their domain of training. All models provide some locally unrealistic predictions which confirms the known poor extrapolation capabilities of neural network models.

5.2. Problem II: uniaxial tension with loading direction change

In this problem, we model the mechanical behavior of both AA5182 and mild steel DC05 because of their different responses when subjected to a loading direction change under uniaxial tension. For each material, the training data is generated by twelve HAH simulations with four distinct levels of strain (0.04, 0.08, 0.12 and 0.16) at the instant of loading direction change, and three different direction changes ($\Delta\alpha = 30^\circ, 60^\circ$ and 90°) after pre-loading along the rolling direction.

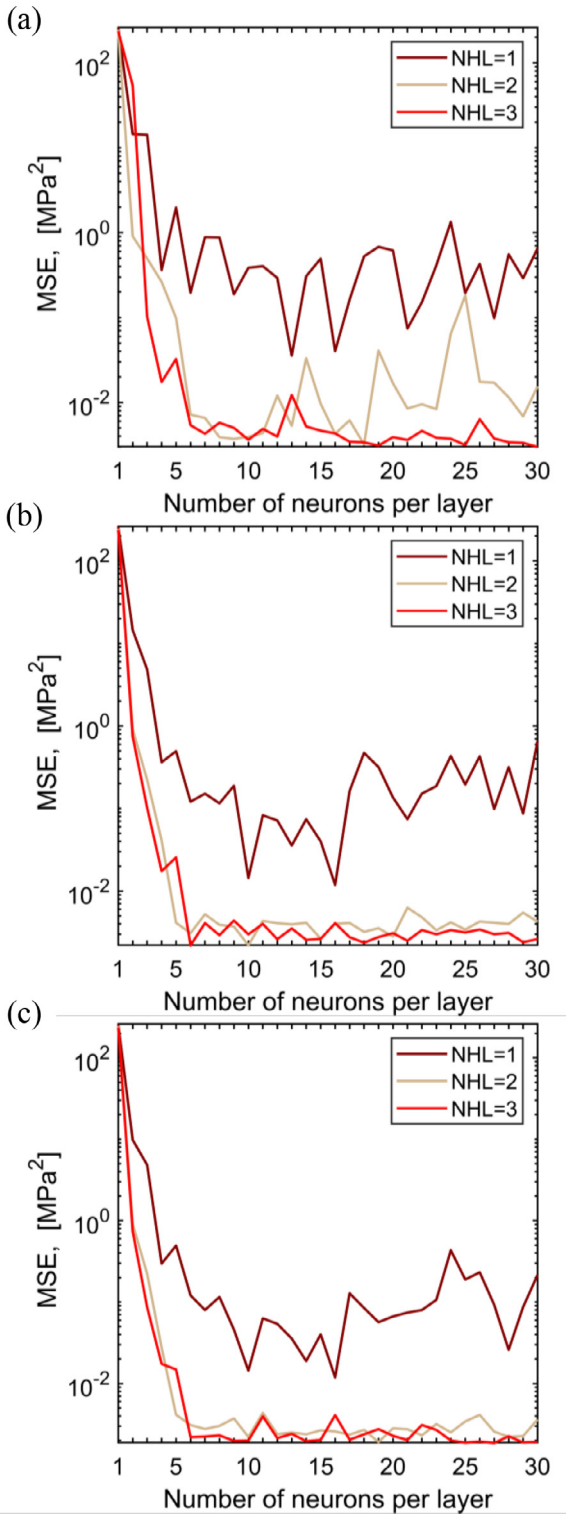


Fig. 3. Results for MSE in $(MPa)^2$ after (a) one, (b) five, and (c) ten training runs as a function of hidden layers (N_{HL}) and neurons per layer (N_{NPL}) .

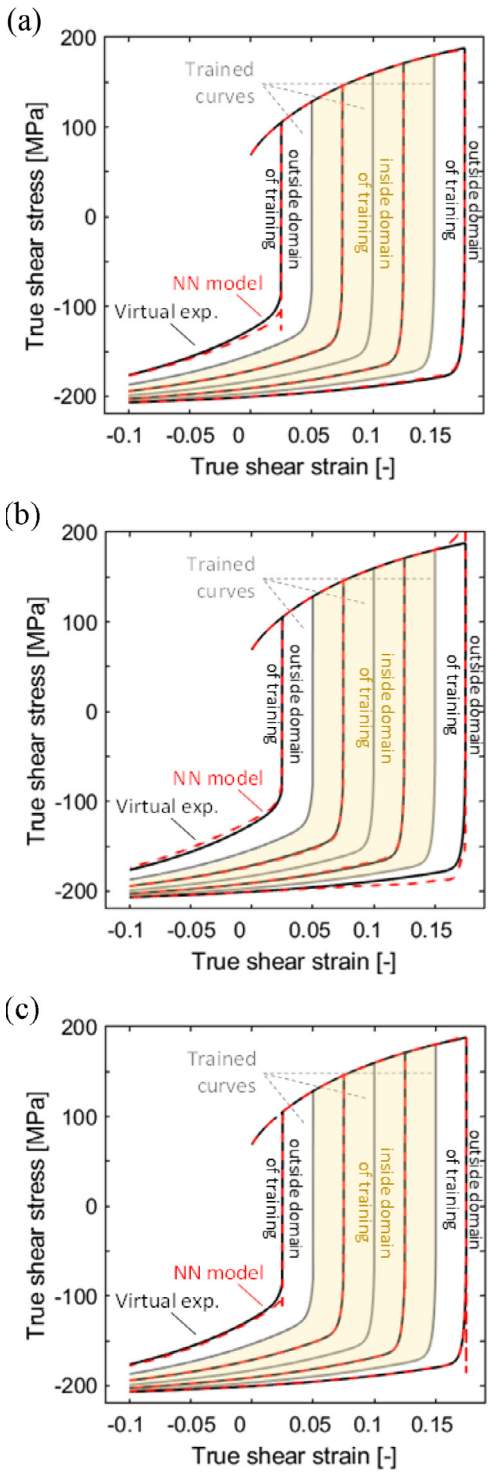


Fig. 4. Results for Problem I: stress-strain curves predicted by three FCNN models (red lines) for reverse loading of AA5182: (a) [2-6-6-1], (b) [2-25-25-25-1] and (c) [2-30-30-30-1] architecture The curves representing training data are shown as gray solid lines. The solid black lines represent the testing data.

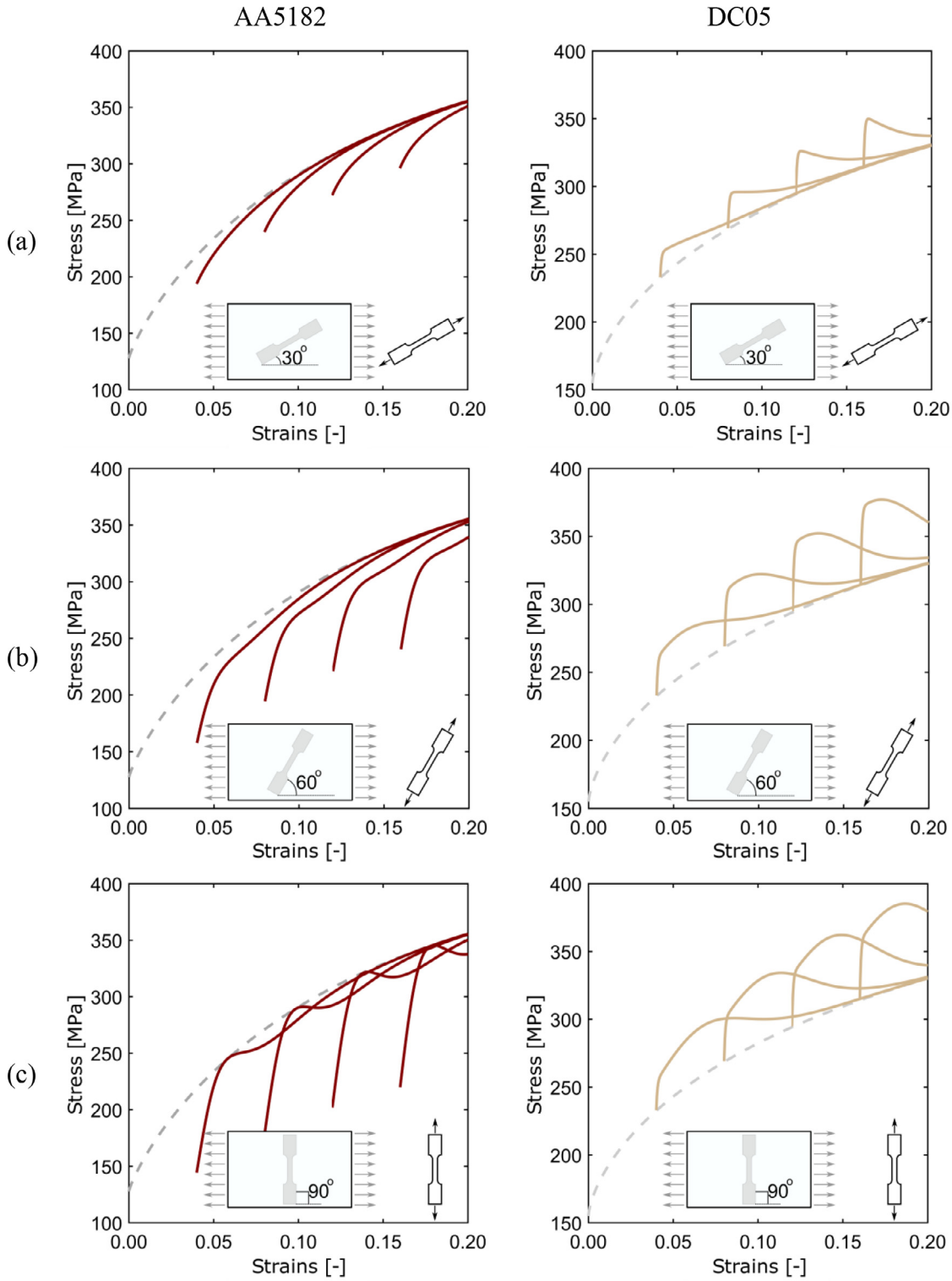


Fig. 5. Training data for Problem II: stress-strain response of AA5182 and DC05 for four different pre-strains (0.04, 0.08, 0.12 and 0.16) at the instant of loading direction change, and three different direction changes (a) 30°, (b) 60°, (c) 90° after pre-loading along the rolling direction.

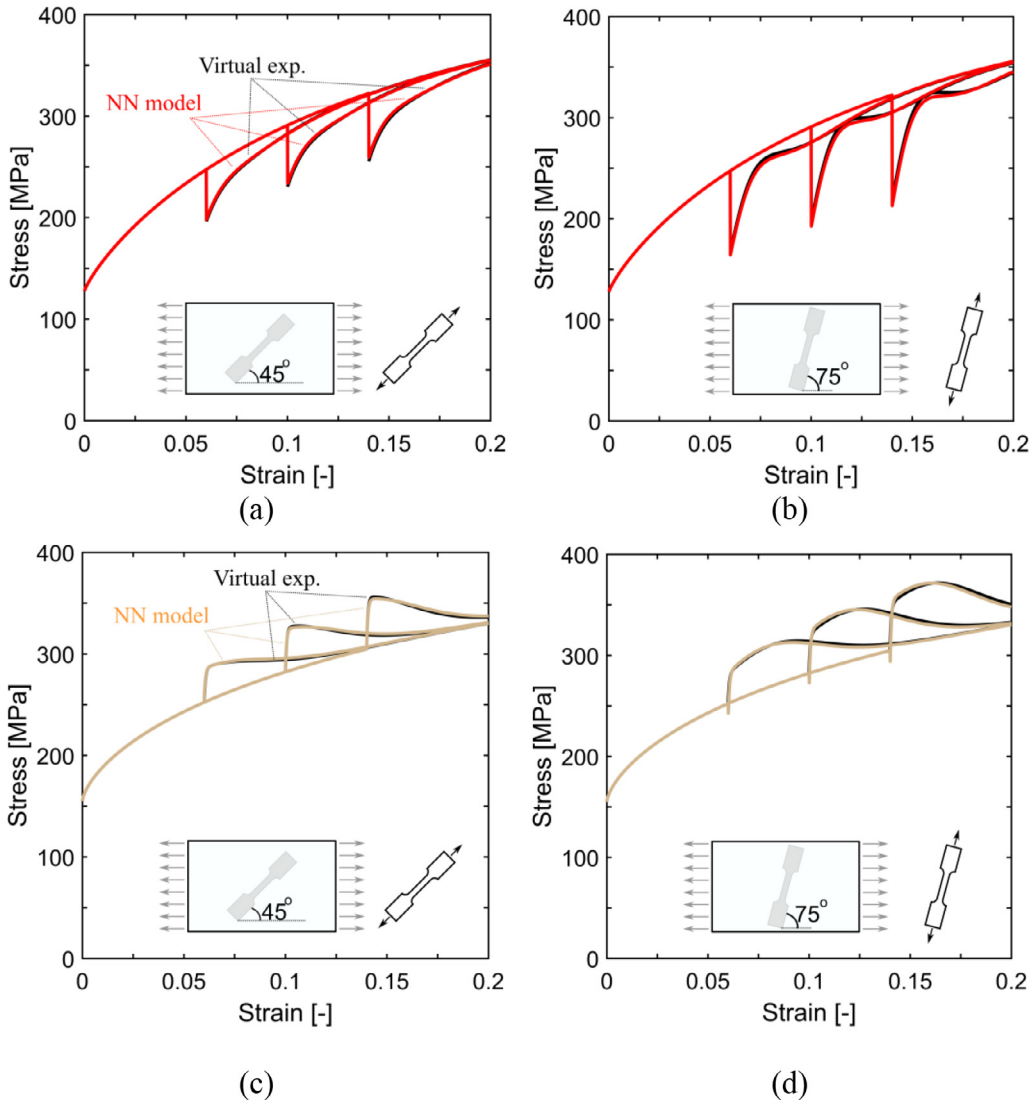


Fig. 6. Results for Problem II (model testing): Stress-strain curve predictions of the neural network for (a-b) AA5182 (red curves), and (c-d) DC05 (orange curves). The solid black lines are generated by the HAH plasticity model (all curves lie approximately on top of each other). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

Fig. 5 displays the stress-strain curves of the loading direction change by 30°, 60°, and 90° with respect to the rolling direction. The AA5182 exhibits latent contraction, which results in re-yielding at a lower stress than that for monotonic loading, whereas the stress overshoots in the case of the DC05 steel. For both materials, the second-stage stress evolution aligns with the monotonic stress-strain curve as the deformation progresses further. These twelve generated logarithmic stress-strain responses for each material (4 different pre-strains $\times$ 3 different loading directions after pre-loading) are then used as a training dataset for machine learning.

The same domain of hyperparameters as for the previous problem is considered, $N_{HL} = \{1, 2, 3\}$ and $N_{NPL} = \{1, 2, .., 30\}$. The identified networks are then tested using a new set of HAH simulations, i.e., three different pre-strains (0.06, 0.10, and 0.14) and different angles (45° and 75°) for the second tensile loading stage. Fig. 6 displays the FCNN approximation for a [3-20-20-20-1] network architecture. The FCNN predictions (represented by colored solid lines) are in excellent agreement with the results from HAH simulations (represented by solid black lines). The network predicts the overshooting in the subsequent second strain path in DC05 steel and is also capable of predicting the latent contraction in aluminum 5182.

5.3. Problem III: monotonic radial loading under plane stress conditions

The plane stress plasticity problem for monotonic radial loading is considered next. Here, the training data is generated through single element simulations for loading trajectories defined by the equations

$$\begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{pmatrix} = r \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\phi \end{pmatrix}. \tag{37}$$

To cover the full sphere in strain space, we limit the coordinates to the intervals $0 \leq \theta < 180^\circ$ for the polar angle and $-180^\circ \leq \phi < 180^\circ$ for the azimuthal angle. We use a 9×18 grid for discretizing the $[\theta, \phi]-$space with a grid spacing of 20°. For a given combination of angles $\{\theta, \phi\}_i$, the HAH simulation is performed by monotonically increasing the radial coordinate from 0 to 0.2. Fig. 7a depicts the resulting 162 loading paths in strain space. Along each path, 272 data points are extracted; 80 points for the first 4% of deformation and 192 points for the remainder. In total, the training data comprises 44064 data points of the type $\{\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}, \sigma_{xx}, \sigma_{yy}, \sigma_{xy}\}_i$.

In close analogy with the previous two problems, the investigation of the hyperparameters revealed that a feedforward network with 3 hidden layers and 20 neurons per layer provides a suitable approximation of all training data. The evolution of the MSE as a function of the number of epochs shows that the training and validation curves are lying almost on top of each other which may be interpreted as good generalization ability of the network model (see Fig. 7b). An example for model testing ($\phi = -160^o$ and $\theta = 33^o$) is shown in Fig. 7c. The HAH curves are represented by solid black lines, while the solid red lines show the FCNN predictions. It confirms that the trained FCNN model is able to predict the change in the shape of the stress-strain curve from the elastic to plastic part (see also zoomed box in Fig. 7c).

5.4. Problem IV: arbitrary loading paths under plane stress conditions

The neural network model of HAH plasticity is further extended by considering arbitrary non-linear loading paths. All loading paths are comprised in a cube Ω in 3D strain space:

$$\Omega: |\varepsilon_{xx}| \leq 0.2, \ |\varepsilon_{yy}| \leq 0.2 \text{ and } |\varepsilon_{xy}| \leq 0.2 \tag{38}$$

They all start and end at the origin and pass through six random points within the cubic domain Ω. After drawing the random points, a smooth strain path passing through all six points is generated using cubic spline interpolation (Fig. 8a). In total, 3000 strain paths are generated using this procedure. The corresponding stress histories are then computed through single element simulations with the HAH model. From each simulation, a sequence of data points of the type $\{\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}, \sigma_{xx}, \sigma_{yy}, \sigma_{xy}\}_{<\tau_i>}$ with $\tau_i = 1, 2, ..., 1000$ is obtained. In sum, the total database for model training and testing comprised three million points. It is split at a ratio of 3:1 into data for training and testing.

Preliminary training runs with different hyperparameters $\{N_{GRU}, N_{NPU}, N_{HL}, N_{NPL}\}$ for the GRU-FCNN model led to configurations featuring one or two GRU cells in series with one or two hidden layers, with both the GRU cells and hidden layers featuring 50 to 100 neurons each. When using two GRUs with $N_{NPU} = 50$, variations of the architecture of the FCNN portion from one to two layers or from 50 to 100 neurons per layer had almost no impact on the MSE over 1000 epochs (Fig. 8b). However, improvements could be achieved when increasing the number of neurons per GRU to $N_{NPU} = 100$ (Fig. 8c), with a final normalized training MSE as low as 1.3×10^{-3} for a network with $\{N_{GRU}, N_{NPU}, N_{HL}, N_{NPL}\} = \{2, 100, 1, 100\}$. The normalized testing MSE for the best model (among those considered) is 1.5×10^{-3}. It features a total of $N_{para} = 101,903$ weights and biases to be learned. The total breaks down to

- 31,200 parameters for the first GRU,
- 60,300 parameters for the second GRU,
- 10,100 parameters for the hidden layer of the FCNN, and
- 303 parameters for the output layer.

The detailed results for a testing case are shown in Fig. 8d-e. The prescribed histories of all components of the strain tensor are plotted in Fig. 8d. It shows a high level of complexity with multiple loading reversals at the component level. In Fig. 8e, the corresponding stress histories predicted by the HAH model are represented by black curves, while red curves show the stress histories predicted by the trained GRU-FCNN model. As expected from the low testing MSE, excellent agreement of all black and red curves is observed which elucidates the high potential of recurrent neural network models for describing the multi-axial stress-strain response of anisotropic elasto-plastic solids.

6. Application: constitutive model for 2D foam

The results obtained for arbitrary loading paths (Problem IV, Section 5.4) are particularly promising as the same neural network modeling framework is expected to apply to any anisotropic elasto-plastic solid provided that the wealth of data required for training can be generated. This is the case whenever virtual experiments can be performed, i.e., if a material model is available at a small length scale. The neural network can then be trained based on the results from the detailed

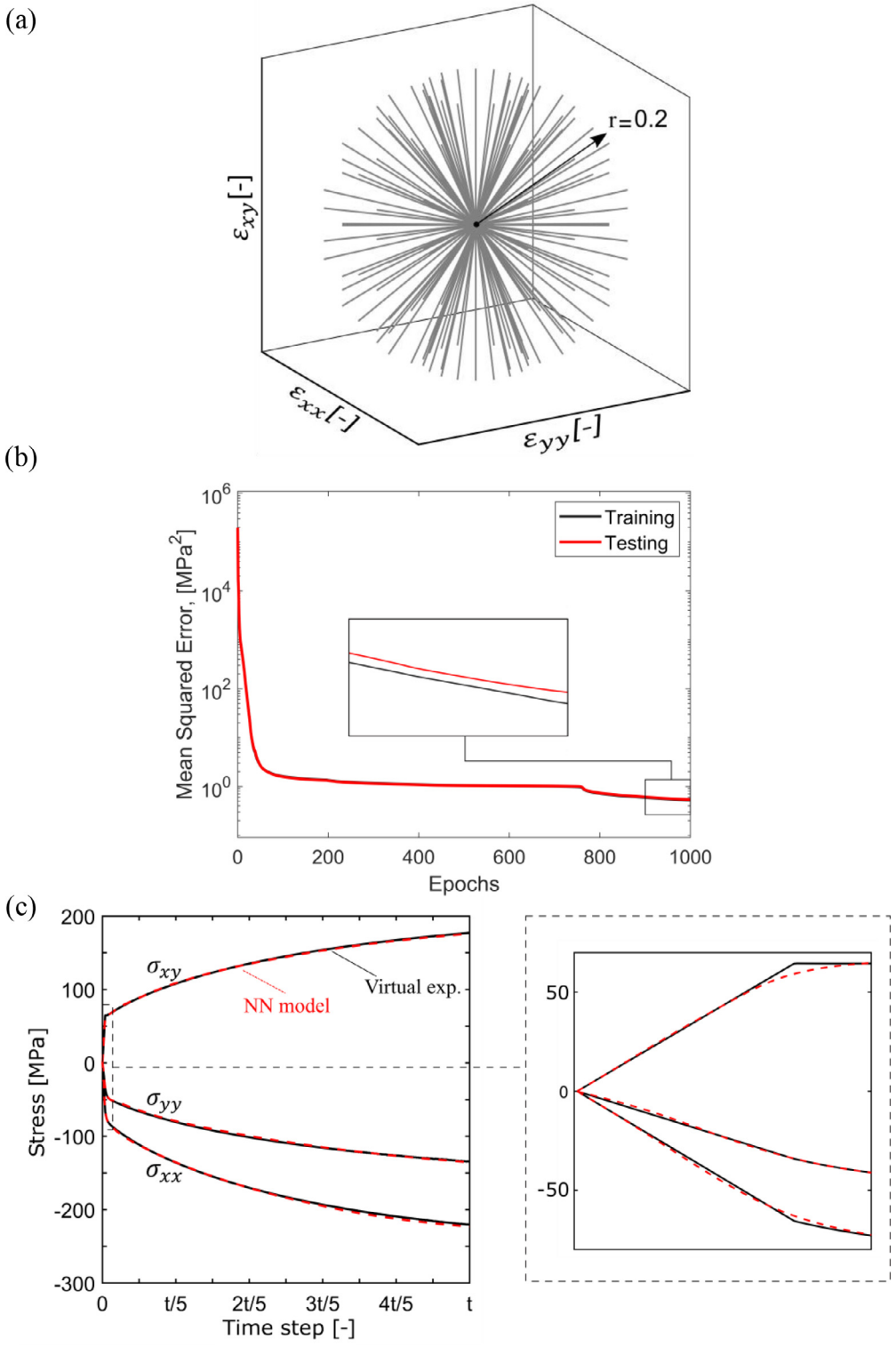


Fig. 7. Results for Problem III: (a) Loading paths in strain space, (b) Training performance of FCNN with [3-20-20-20-3] architecture, (c) Testing of the FCNN model for monotonic radial loading with $\phi = -160^o$ and $\theta = 33^o$.

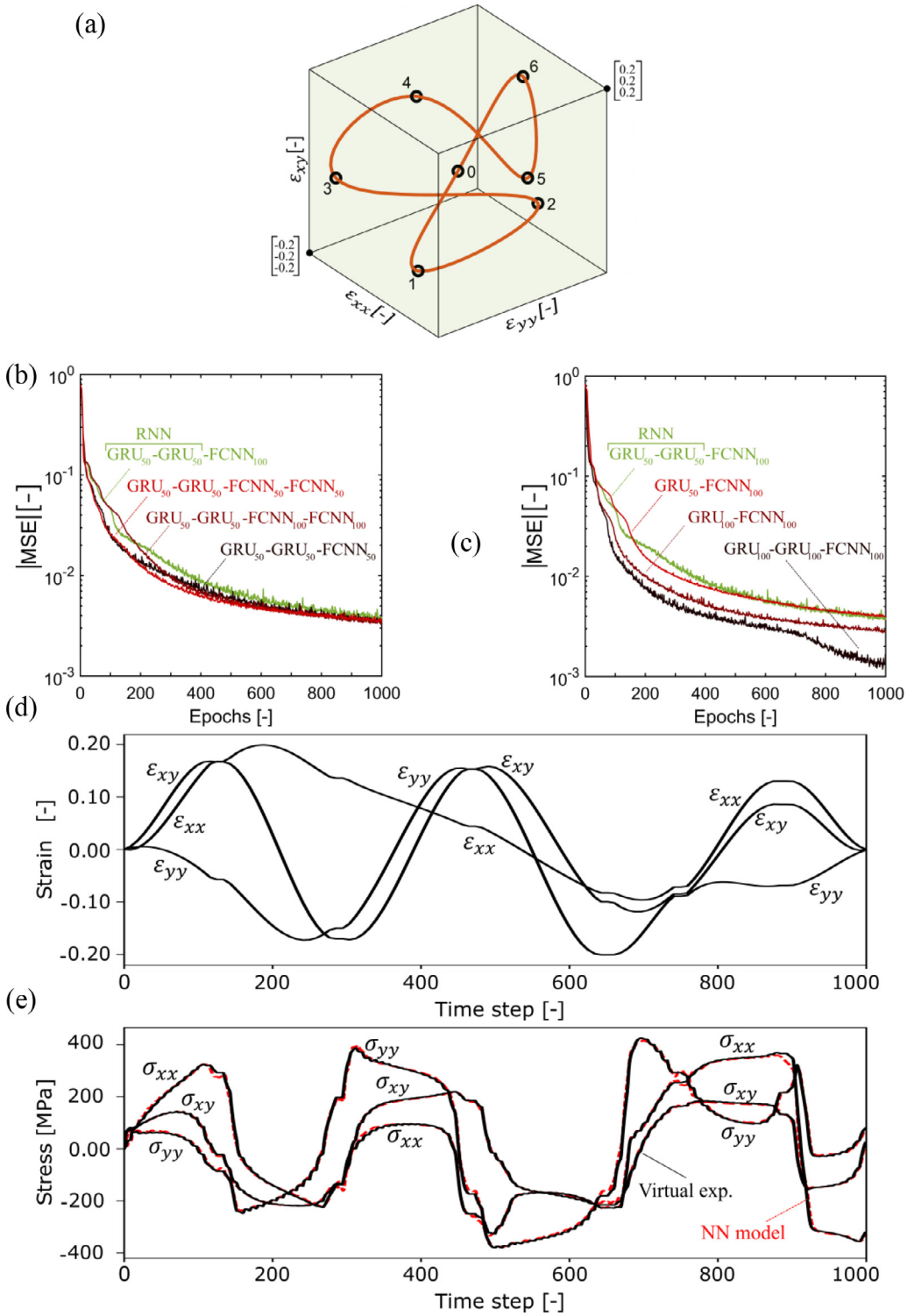


Fig. 8. Results for Problem IV: (a) Example of an arbitrary loading path generated by 3D spline curves in strain space, (b-c) training performance of GRU-FCNN models with different hyperparameters (d) strain histories for model testing; (e) stress histories predicted by HAH model (black curves) and of GRU100-FCNN100 model (red curves). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

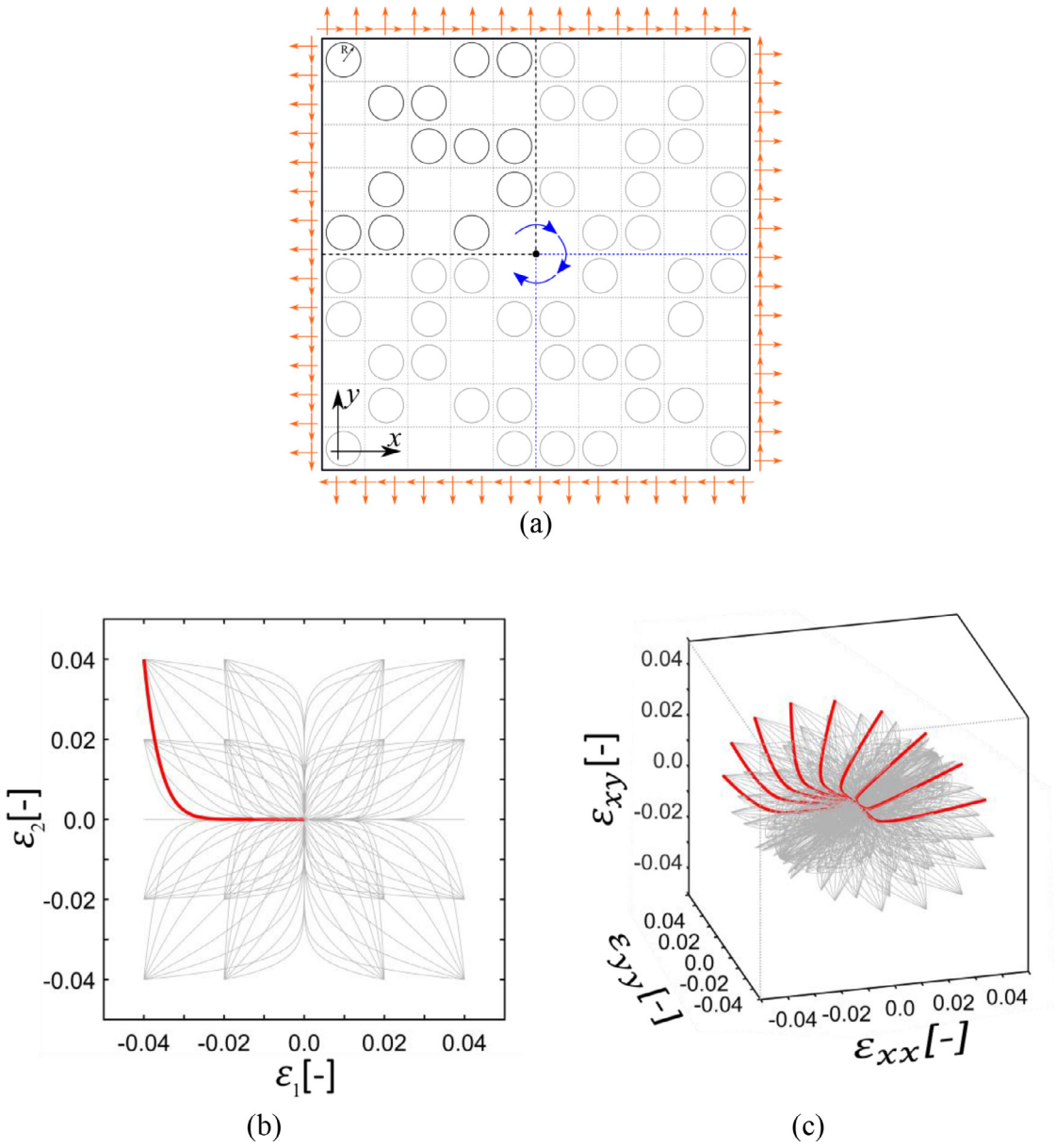


Fig. 9. (a) Unit cell of two-dimensional foam structure with the randomly-distributed holes. Generation of loading paths for training (a) in 2D, and (c) in 3D strain space.

model and serve as a computationally-efficient counterpart for modeling the material response at the macroscopic level. To demonstrate the applicability of this approach, a neural network model describing the biaxial stress-strain response of a two-dimensional foam is developed.

6.1. Virtual experiments

6.1.1. Detailed FE model of a unit cell

The two-dimensional foam resembles a perforated plate with randomly-distributed holes (Fig. 9a). After distributing 13 holes onto a 5 × 5 square grid, a 10 × 10 grid is populated through successive 90° rotations of the 5 × 5 grid. The hole radius is chosen such that 50% of the area on an elementary square is perforated. The relative density of the foam thus equates to $\rho* = 0.74$, i.e. the porosity is only 26% which corresponds to high-density foam. The solid matrix is assumed to be made from aluminum 5182.

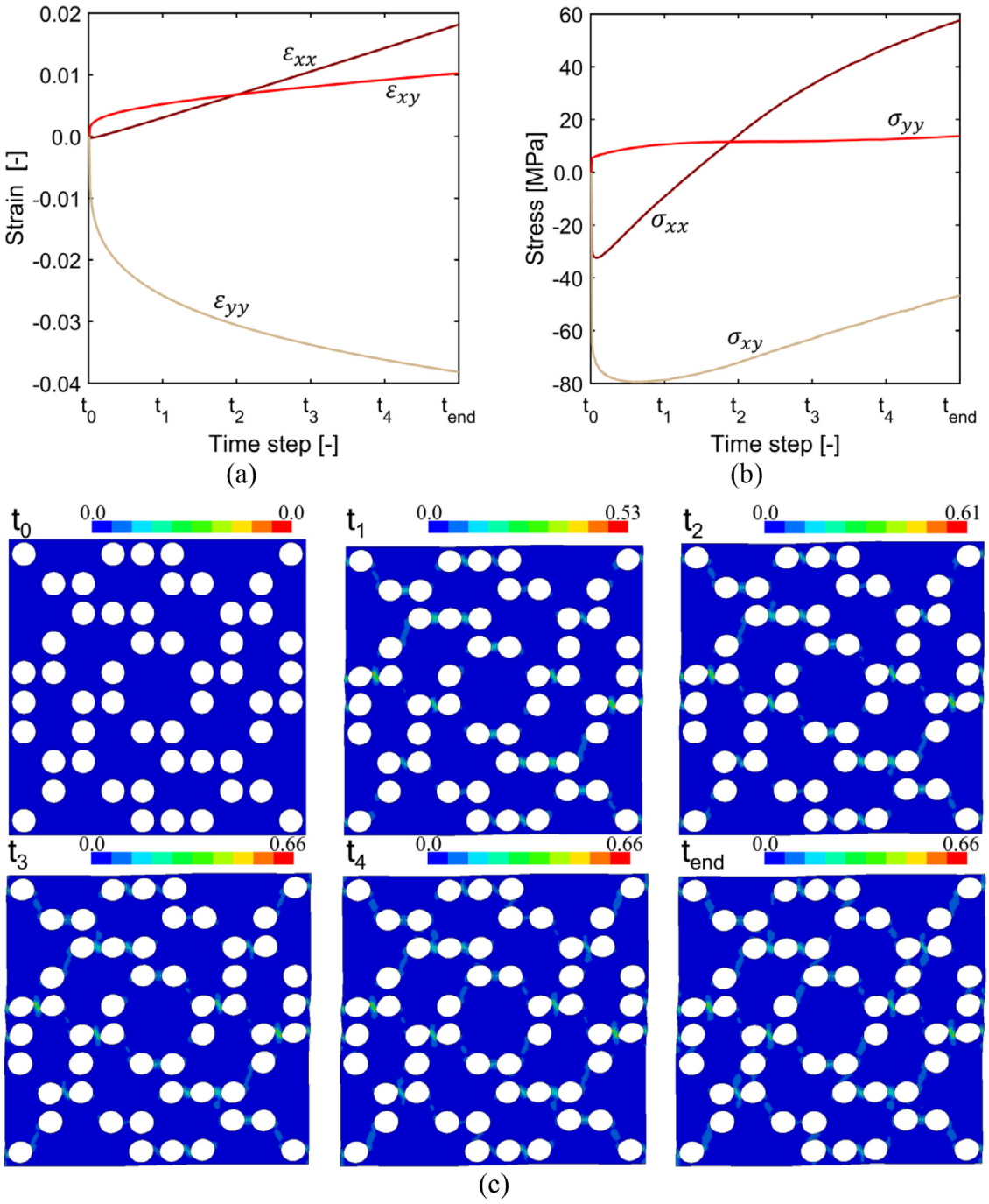


Fig. 10. Results from a representative virtual experiment (unit cell simulation): (a) imposed strain histories, and (b) computed stress histories. (c) Deformed unit cells for selected time steps. The contour plot shows the equivalent plastic strain

To determine the macroscopic stress-strain response, the solid phase of the unit cell is discretized using 40,000 first-order plane-stress elements which corresponds to an element edge length-to-hole diameter ratio of about 0.04. The matrix material is modeled using the anisotropic Yld2000-2d model with HAH hardening. Periodic displacement conditions are enforced through linear kinematic constraints between nodes of opposite faces. The constraints are formulated such that the history of the components of the macroscopic deformation gradient can be prescribed,

$$\mathbf{F}[\mathrm{t}] = F_{xx}[\mathrm{t}]\mathbf{e}_x \otimes \mathbf{e}_x + F_{xy}[\mathrm{t}](\mathbf{e}_x \otimes \mathbf{e}_y) + F_{yx}[\mathrm{t}](\mathbf{e}_y \otimes \mathbf{e}_x) + F_{yy}[\mathrm{t}]\mathbf{e}_y \otimes \mathbf{e}_y, \tag{39}$$

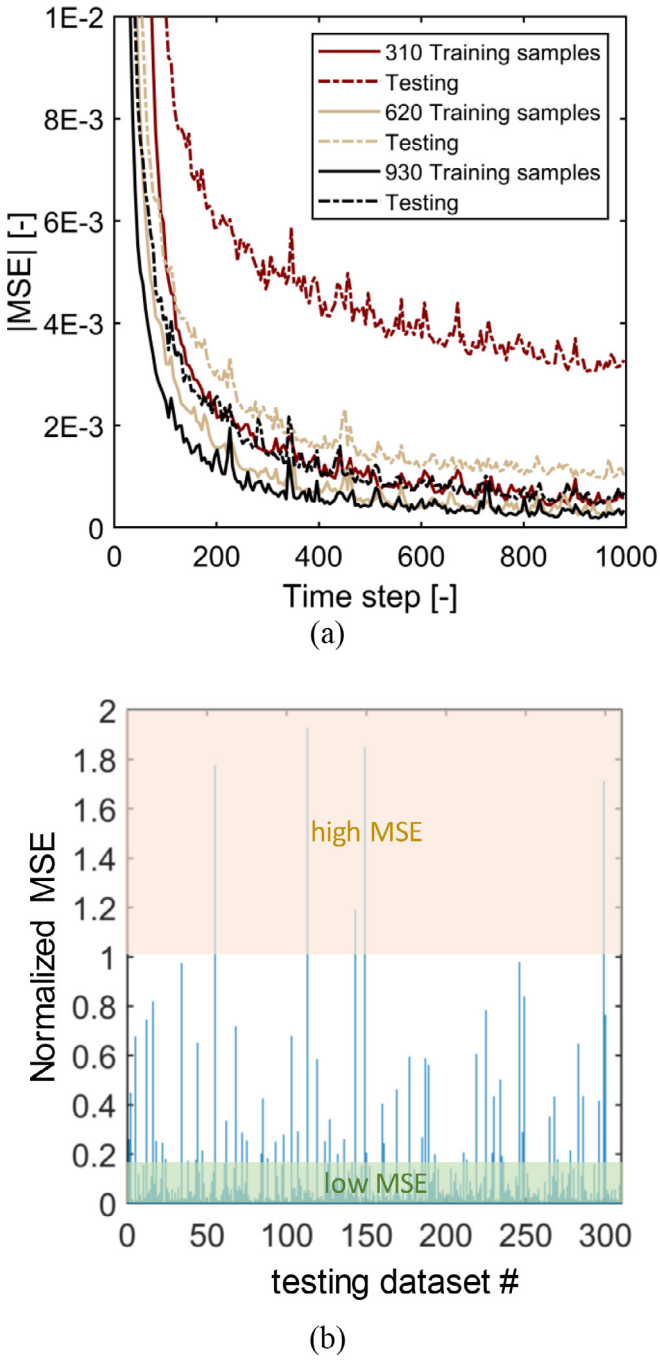


Fig. 11. Training and testing performance of GRU-FCNN models: (a) sensitivity of MSE on size of training data set, (b) distribution of MSE among 310 testing datasets (when using 930 training sets).

while the components of the macroscopic first Piola-Kirchhoff stress tensor **P** (in the x-y-coordinate system) are determined from the reaction forces at the control nodes. The neural network model is formulated in terms of the components of the logarithmic strain tensor ε and the Cauchy stress tensor $\boldsymbol{\sigma}$. They are obtained through the standard transformation relationships

$$\boldsymbol{\varepsilon} = \ln \mathbf{U} \text{ with } \boldsymbol{U}^2 = \boldsymbol{F}^T \boldsymbol{F} \tag{40}$$

and

$$\boldsymbol{\sigma} = (det(\boldsymbol{F}))^{-1} \boldsymbol{P} \boldsymbol{F}^T. \tag{41}$$

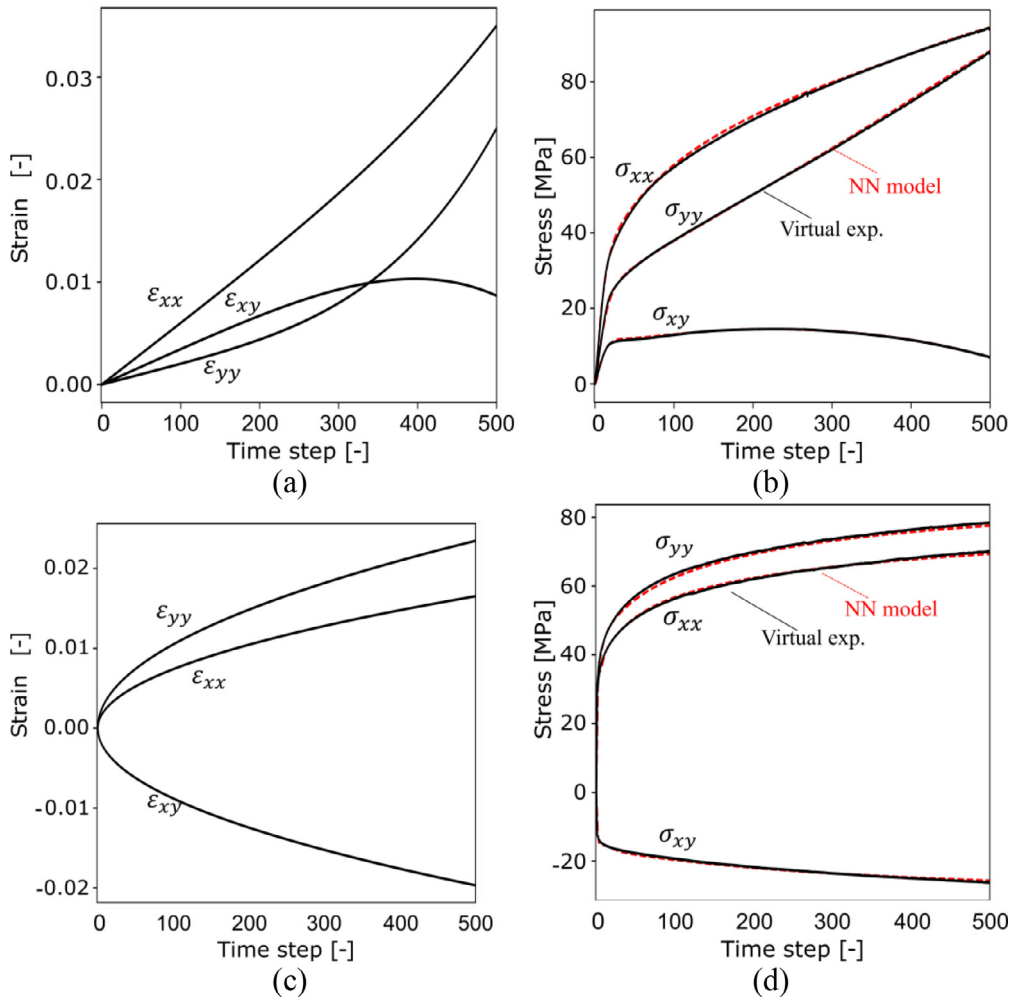


Fig. 12. Testing of the trained GRU100-FCNN100 model for 2D foam – examples with a low MSE: the left column figures show the imposed strain histories, while the right column figures show the corresponding stress histories predicted by the unit cell model (black curves) and the neural network model (red lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

Furthermore, we impose $F_{xy} = F_{yx}$ which implies $\boldsymbol{U} = \boldsymbol{F}$. As a consequence, the deformation gradient history can also be expressed as a function of the logarithmic strain history. More specifically, by introducing the angle α between the direction of the first principal strain ε_1 and the $\mathbf{e}_x$-direction, the deformation gradient in the x-y coordinate system reads

$$\{\mathbf{F}\}_{xy} = \exp(\varepsilon_1)\begin{pmatrix}\cos\alpha\\ \sin\alpha\end{pmatrix}\otimes\begin{pmatrix}\cos\alpha\\ \sin\alpha\end{pmatrix} + \exp(\varepsilon_2)\begin{pmatrix}-\sin\alpha\\ \cos\alpha\end{pmatrix}\otimes\begin{pmatrix}-\sin\alpha\\ \cos\alpha\end{pmatrix}, \tag{42}$$

where ε_2 denotes the second logarithmic principal strain (which always acts along an in-plane direction perpendicular to that of ε_1).

6.2. Virtual experiments

The first set of 140 loading paths is generated in the ε_1 versus ε_2 plane. These are based on power-functions that are linearly transformed to attain one of the 16 grid points with the coordinates $\varepsilon_1^e = \{-0.04, -0.02, 0, 0.02, 0.04\}$ and $\varepsilon_2^e = \{-0.04, -0.02, 0, 0.02, 0.04\}$

$$\varepsilon_2 = \varepsilon_2^e\left(\frac{\varepsilon_1}{\varepsilon_1^e}\right)^n \tag{43}$$

with the exponent n ∈ [0.1, 0.25, 0.5, 1, 2, 4, 10], see Fig. 9b. The number of loading paths in strain space is then further increased to 1260 by varying the direction of the first principal strain in 10°-increments from 0 to 80°. The resulting strain

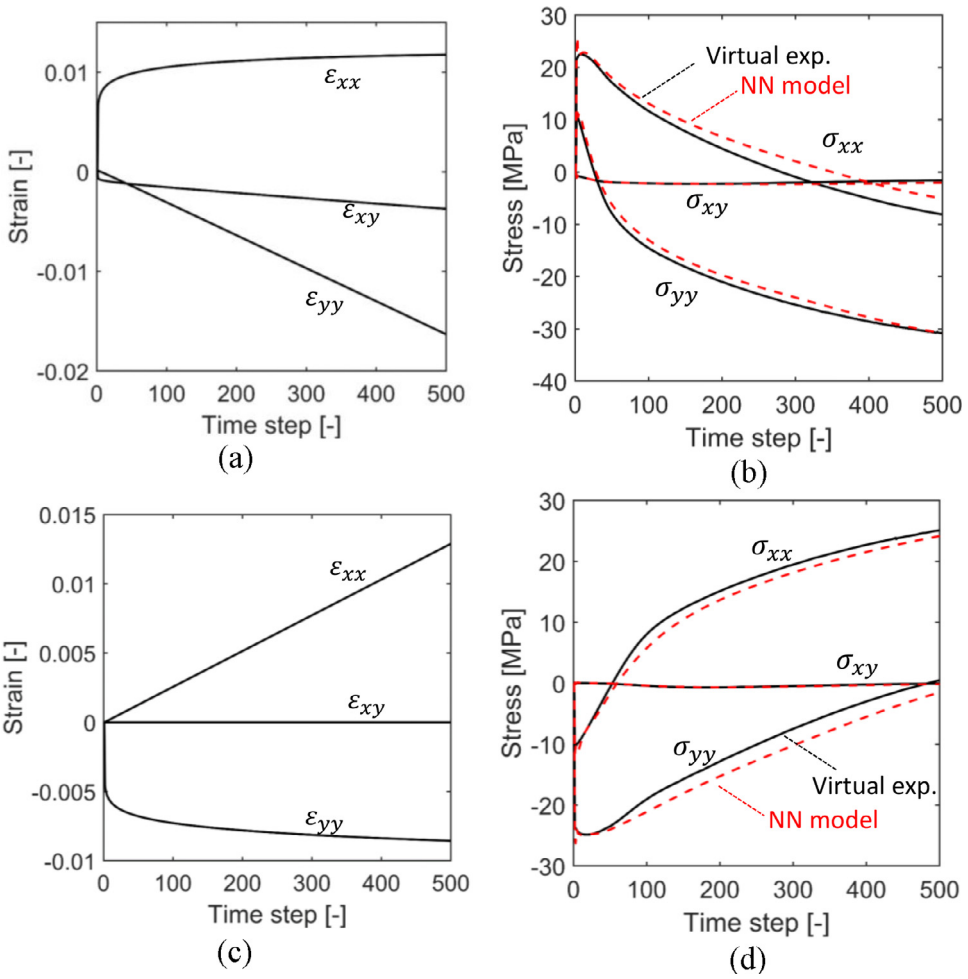


Fig. 13. Testing of the trained GRU100-FCNN100 model for 2D foam – examples with a high MSE: the left column figures show the imposed strain histories, while the right column figures show the corresponding stress histories predicted by the unit cell model (black curves) and the neural network model (red lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

Table 2
The hardening parameters of the studied alloys (Peters, 2015).

	σ_y [MPa]	σ_{sat} [MPa]	m [-]	n [-]
AA5182	434.28	127.63	5.26	0.84
DC05	451.89	154.95	2.62	0.67

paths in the 3D strain space $\{\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}\}$ are depicted in Fig. 9c. For reference, the nine paths corresponding to the red path shown in Fig. 9b are also highlighted in red in Fig. 9c.

The results from a representative virtual experiment are shown in Fig. 10. Plot (a) depicts the strain inputs, while plot (b) illustrates the resulting stresses based on the forces in the x- and y-directions on the control points. Even though the input strains are based on monotonic functions, the computed histories of the stresses are not necessarily monotonic. For instance, the stress component σ_{xx} in Fig. 10b starts out as compression and changes to tension after approximately one-third of the deformation has been applied to the unit cell. The contour-plots in Fig. 10c depict the equivalent plastic strain distribution at six different stages of loading. As expected, we observe strain concentrations near the holes that lead to the localization of plastic deformation at the ligament between neighboring holes.

6.3. Neural network-based constitutive model

The same neural network architecture as the one presented for the arbitrary loading paths (see Section 5.4) is used to model the large deformation response of the high-density aluminum foam: the GRU-FCNN model with the hyperparameters

Table 3
Yld2000-2d parameters of the studied alloys.

	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	q
AA5182	0.978	1.021	1.290	1.081	1.064	1.234	1.015	0.912	8
DC05	1.084	0.982	0.846	0.881	0.908	0.835	0.972	0.975	6

Table 4
HAH model parameters of studied alloys to predict the Bauschinger effect (Peters, 2015).

	k_1	k_2	k_3	k_4	k_5
AA5182	250	10.0	0.10	0.90	10.0
DC05	158.37	6.33	0.10	0.92	10.0

Table 5
HAH model parameters of studied alloys to predict the latent effects (Peters, 2015).

	k	L	k_L	C	k_C
AA5182	30.0	1.0	0.0	1.35	100.0
DC05	28.0	1.86	1600.0	1.0	0.0

$N_{GRU} = 2$ and $N_{NPU} = 100$ for the GRU part, and $N_{HL} = 1$ and $N_{NPL} = 100$ for the FCNN part. The results for each of the 1240 loading paths are represented by a sequence of data points of the type $\{\underbrace{\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}}_{N_{IN}=3}, \underbrace{\sigma_{xx}, \sigma_{yy}, \sigma_{xy}, \varepsilon_{xy}}_{N_{OUT}=4}\}\langle\tau_i\rangle$ with $\tau_i = 1, 2, ..., 500$. In total, the set of data comprises 630,000 data points. The data is randomly split at a 3:1 ratio for training and testing. In other words, 930 loading paths out of 1240 generated ones are used to train the neural network, while the remainder 310 loading paths are used for testing. The network is trained for 1000 epochs which led to MSEs (for the normalized outputs) of $\mathbf{3.3}6 \times 10^{-\mathbf{4}}$ and $6.7 \times 10^{-\mathbf{4}}$ for training and testing, respectively (Fig. 11a). Fig. 11b shows the relative MSE for each of the 310 testing datasets. The plot reveals substantial variations among the testing cases. To assess the accuracy from an engineering perspective, we plotted the corresponding stress evolutions for selected examples with low and high MSEs in Figs. 13 and 14, respectively. The left column plots depict the prescribed strain histories, while the desired output histories (as predicted through unit cell analysis) are represented by black curves in the right column plots. The red curves show the predictions of the GRU-FCNN model. For the low MSE cases (Fig. 12), the black and red curves lie almost perfectly on top of each other. For the high MSE cases (Fig. 13), the neural network model also provides reasonable predictions. The dashed red curves are smooth and follow the results from the virtual experiments with an absolute error of less than 3*MPa*. It is thus tentatively concluded that the proposed neural network model is able to describe the highly non-linear stress-strain response of a two-dimensional metallic foam.

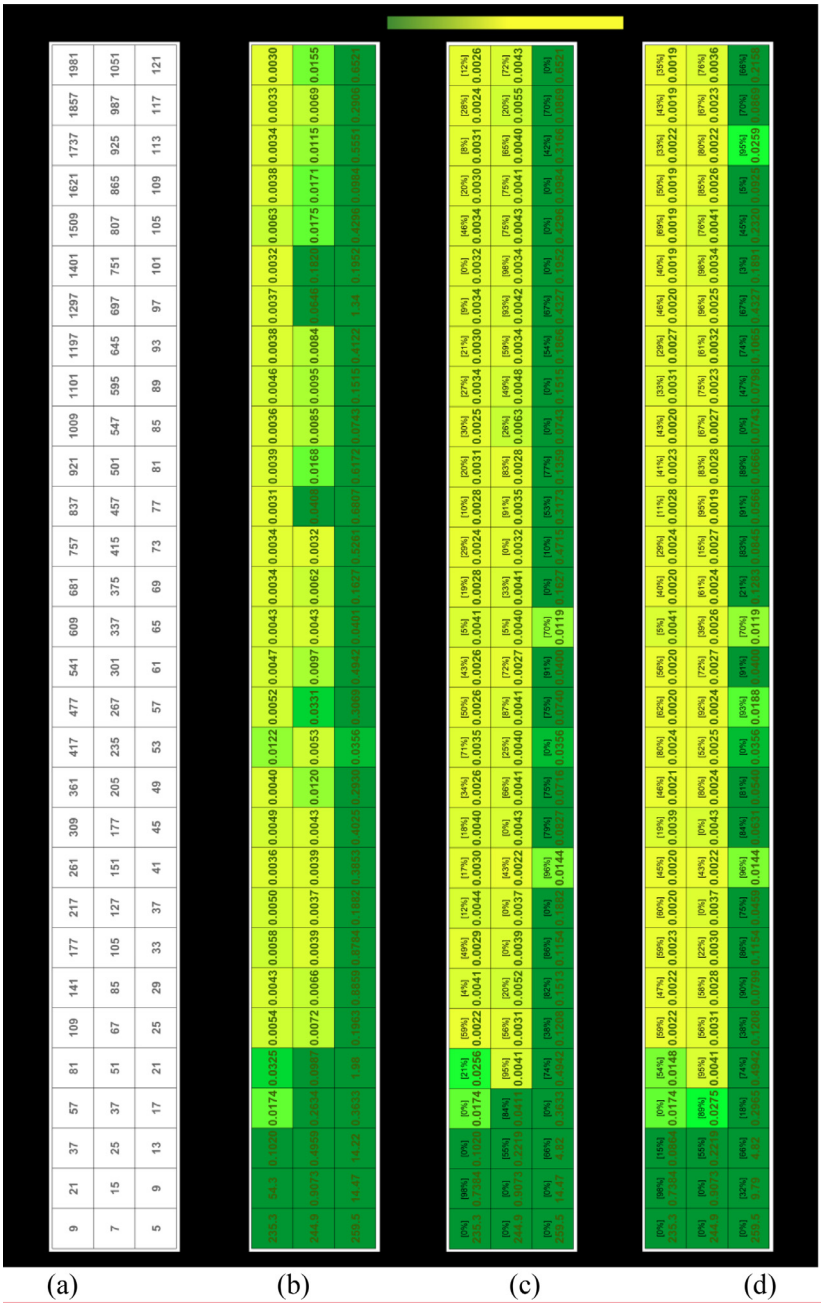
To reveal the sensitivity of the network accuracy on the size of the training set, we also repeated the training for the training-to-testing data split ratios 1:1 (620 curves for training) and 1:3 (310 curves for training). The corresponding training performance curves are shown in Fig. 11 next to the reults for a 3:1 split (930 curves for training). A significant increase in testing performance is observed when increasing the number of training samples from 310 to 620. The subsequent increase from 620 to 930 is also beneficial, but less significant.

6.4. Discussion

The present results elucidate the potential of neural network based models as an alternative to differential equation based constitutive models for describing the elasto-plastic response of solids. It is well-known through the universal approximation theorem (Cybenko, 1989) that neural networks can represent any smooth non-linear function. It is difficult to demonstate that the incremental plasticity problem at hand can be described by a single non-linear function. It was thus an important first step to generate empirical evidence that suggest that GRU based networks are able to learn the mean features of the anisotropic Yld2000 plasticity model with HAH hardening.

Many questions are still left open after completing the present work. In view of modeling real engineering materials it is important to come up with machine learning based models that can be identified from experiments. The brief sensitivity analysis of the effect of the size of th training set clearly indicated that the more data the better the model accuracy. This implies that the introduction of neural network based constitutive models would also require a paradigm shift in experimental mechanics, where stress-strain curves would need to be measured for hundreds if not thousands of different multi-axial loading paths. This is certainly beyond today's testing capabilities, but the technology to achieve this may be found in other areas of mechanical engineering such as robotics and automation.

Table 6
Results for Problem I: (a) Number of parameters in FCNN as a function of hidden layers (N_{HL}) and neurons per layer (N_{NPL}). MSE in $(MPa)^2$ after (b) one, (c) five, and (d) ten training runs. The percentage reported in (c) and (d) corresponds to the improvement compared to (b).



On the computational end, it may also be of interest to compare the computational efficiency of neural network based models with conventional return mapping schemes. Due to the feed forward nature of the neural network models, it is possible to perform a stress update without any iterations. While it is straightforward to implement a neural network model into a modern FE software through the user material subroutine interface, the full computational benefits of a neural network based constitutve model may only become appararent after the professional implementation with optimal parallel computing for solving large structural problems.

From a modeling perspective, other neural network approaches would also need to be explored in the field of plasticity. Next to GRUs, the potential of LSTMs and other RNN solutions would need to be asseesed. It is also crucial to come up

with model testing schemes that guarantee the unconditional robustness of the neural networks for arbitrary loading paths in strain space. Due to the known capababities of convolutional neural networks (CNNs) for recognizing image features, it will aslo be of interest to build neural network based constitutive models directly based on microstructural images (such as EBSD pictures of polycrystals or the pore distribution in foams).

7. Conclusions

It has been demonstrated that deep learning with neural networks provides a very powerful modeling framework that is suitable for data-driven constitutive modeling. This strength of Fully-Connected Neural Networks (FCNNs) has been shown in the context of problems dealing with the description of the uniaxial stress-strain response of elasto-plastic solids. Even rather complex phenomena such as the Bauschinger effect and hardening stagnation can be successfully described. For these types of modeling problems, the main challenge is to choose the input variables such that the mechanical problem can be represented by a non-linear function.

In the multi-axial case, it is far more challenging to reduce plasticity problems to a single non-linear function. In conventional physics-based plasticity models, the loading path dependence of the stress is captured by the evolution of internal variables such as the plastic strains and back stresses. In the present work, the anisotropic plane stress plasticity has been modeled using Recurrent Neural Networks (RNNs). RNNs are equipped with memory which enables them to account for the loading path history when making predictions. In a first step, we have demonstrated that RNNs can replicate the predictions for arbitrary loading paths of one of the most complex anisotropic plasticity models (Yld2000-2d with HAH hardening). In a second step, the deformation response of a two-dimensional metallic foam has been successfully described through an RNN model. It is concluded that RNNs provide a data-driven modeling framework, which is particularly suitable for modeling the plastic response of materials provided that a wealth of training data can be generated by means of unit cell or RVE simulations (e.g. composites, polycrystals, architected materials).

Neural network based constitutive models will gain even more importance (1) if it can be demonstrated that they are faster than conventional return mapping schemes, and (2) if experimental mechanics can provide the stress-strain response of materials for hundreds of different multiaxial loading paths. Both conditions will probably be fulfilled in the near future which supports the development of specific neural network architectures to describe the stress-strain response of elasto-plastic solids.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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